Manual

AIP Functions Shop Floor Data / Machine Data

AIP-BMD 8.2

Version 1.1.23167

Last changed on: 08.09.2020

Copyright

Contents

1 AIP Functions - Shop Floor Data/Machine Data

Purpose

The AIP features contained in this function package make it possible to enter data directly in production using shop floor terminals or shop floor PCs.

Implementation considerations

You use the function package if you:

- want to know the current status of each order/operation
- want to know about the progress of the order (time-related and/or quantity-based)
- want to record the activities performed (machine time, labor utilization, quantities) in order to
  - evaluate them, or
  - upload them to a higher-level ERP system.
- would like to monitor machines or would like to enter or track the current status of each machine.
- want to electronically move information such as documents or production notes into Production.

Integration

AIP offers the ability to automatically transfer data from the machine, to record it and to forward it to the server to be posted. Various interfaces are available for this purpose.

Data entered using AIP can be displayed or evaluated in various applications in MOC.

Features

Order-related data entry and posting functions

- Posting functions
  - Log on/log off, interruption and partial confirmation/upload of operations
- Entry of quantities
  - Entry of yield and scrap quantities
  - Entry of rework quantities or open quantities (customizing)
- Recording scrap reasons
  - Entry of scrap reasons
- Reasons for deviation

  - Entry of reasons for deviation in the event of over or underdelivery
- Displaying lists / icons
  - Basic display of workplaces/machines assigned to the terminal or operations logged onto a selected workplace. Presentation in the form of a list or in icon view
- Display of order backlog data
  - Display of selected operation-related information about planned operations or about currently logged on operations: Order backlog data, material components, production resources and tools
- Display of machine data
  - Display of machine master data and shift-related quantities
- Display notes
  - Display of operation-related notes entered on the client
- Modify default values
  - User dialog used to modify the default value for "target quantity"
- Entry of BDE comments
  - Entry of OP-related comments (free texts) added to any particularities documented during the course of the order
- Status log
  - Display of operation or personnel postings made during the current shift

Personnel-related entry and posting functions

- Personnel logon/logoff
  - Personnel logons and logoffs to and from operations or the workplace

Machine monitoring and status recording

- Cycle/operating signal monitoring
  - Automatic monitoring of the production status of machines based on cycle signals or operating signals (requires a connection from the machine to the shop floor terminal).
- Malfunction reason
  - Dialog in which to enter reasons for a disturbance detected by the machine monitoring function
- Modify default values
  - User dialog used to modify the default value for "target cycle" needed for the cycle-based machine monitoring function.
- Manual entry of machine status
  - Manual entry of machine states/statuses
- Status log

- Display of the machine states/statuses posted during the current shift. Function for the subsequent assignment of reasons to disturbances/malfunctions that had not yet been given one.

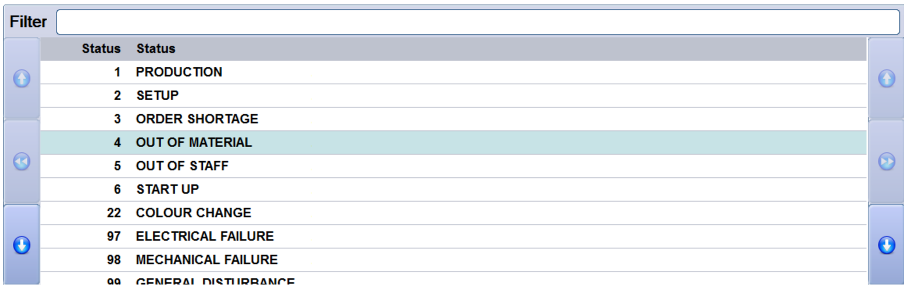
Additional licenses might be needed in order to use the functions listed above.

2 AIP2 Operation

2.1 Special Control and Display Elements on the AIP2

Tables

Uniform selection lists are used in AIP 8.2 posting dialogs:



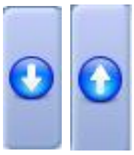
Page 1 • Page 2 If information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. If the user clicks/touches a page, the display directly changes to this page.

If more pages are available than the page numbers displayed, the following buttons can be displayed on the left or right hand side depending on the context (available as of SP10/2016):

- »»» : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 1 ... Page 9* were displayed for the page navigation, the system jumps to *Page 10*.
- ««« : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 10 ... Page* 18 were displayed for the page navigation, the system jumps to *Page* 9.
- : If you click this button, the system directly jumps to *Page 1*.
- : If you click this button, the system directly jumps to the last page.

You can select an operation using the mouse, touch screen, keyboard (arrow keys:'↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Example: When you log on an operation, those operations are available that are included in the sequencing list or planned for the respective workplace or group. When you interrupt an operation, only running operations are available for selection.

 Scrolling page by page (up or down) in the table.

 Scrolling to the left or right. Only those buttons are activated that make sense for the current situation (context sensitive). This figure shows that scrolling to the left has been deactivated.

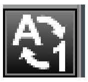
Optionally you can display a “table filter” (customization). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is *case-insensitive*.

Virtual keyboard

Using the virtual keyboard, you can enter data manually via touch screen or a connected mouse. The virtual keyboard is displayed automatically as soon as the focus is on an input field. The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard.

 Moving the virtual keyboard

 Hiding the keyboard for 10 seconds

 Switching between the alphanumeric and numeric keyboard

 Selecting the keyboard layout (language)

 Changing the scaling/size of the keyboard

 To move the keyboard, you must configure the driver accordingly (configuration in the control panel of the terminal/PC)!

If you do not want to display the virtual keyboard in general, you must enter the parameter –t in the entry *parameters*= of the configuration file *ctaip.ini.*

Date display

AIP supports a country-specific date format in dynamic dialogs. The option "short date" has to be selected in the "regional settings" of the Windows "control panel" of the terminal/PC. Please note:

- Years are always four characters long.
- Months and days are always 2 characters long.
- Allowed separators are: '-‘ (minus), '/‘ (backslash) and '.‘ (dot).
- Blanks must not be included in the “short date” format, i.e. the <BLANK> separator is not allowed.
- The date separator “.” (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- English(USA) MM/DD/YYYY
- Danish MM-DD-YYYY
- Customer-specific 1 YYYY-MM-DD
- Customer-specific 2 YYYY/MM/DD

- Customer-specific 3 MM/YYYY/DD

Note

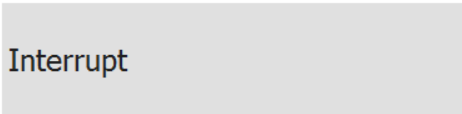
If the date format used is other than the permitted formats, a note appears when the program is started and the date format is set to MM/DD/YYYY.

In the status bar, the year format is shortened and displayed only with two characters.

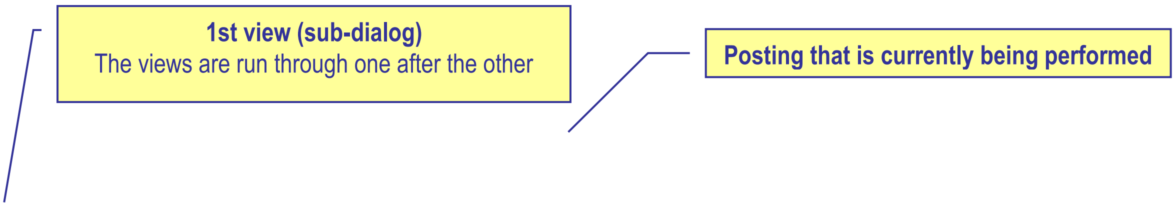
2.2 General description of the posting process on the AIP2

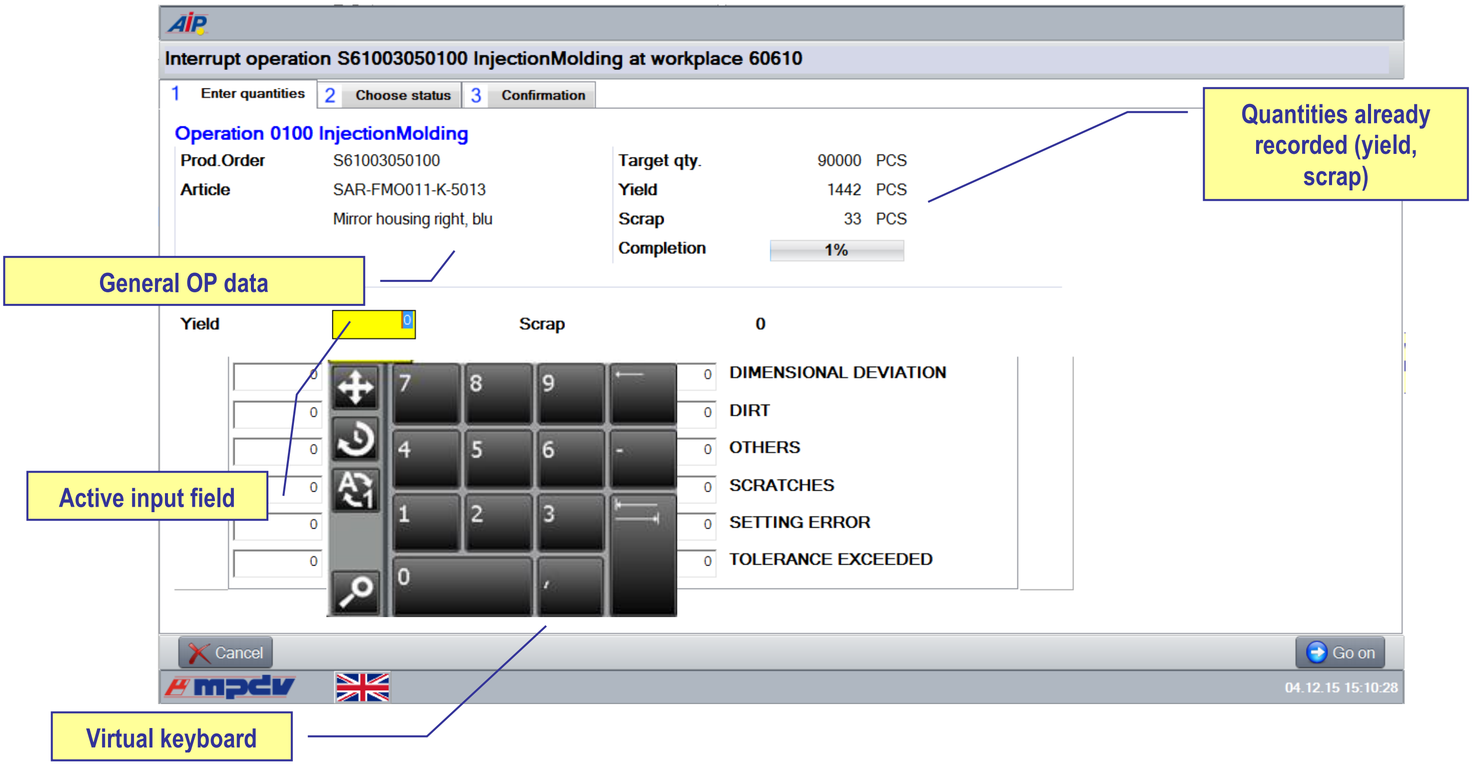
Many AIP posting dialogs are divided into several views (sub-dialogs). These views (sub-dialogs) cover the entire screen so that only one dialog is visible at a time. In a "workflow concept" the user is navigated through the posting dialog step by step. In the following, this process is described using the example *Interrupt operation*. Other posting dialogs are operated in the same way.

The action *Interrupt operation* is performed. To start this action, you click the button *Interrupt* when you have selected an operation:



The dialog *Interrupt operation* opens and the first view (sub-dialog) is displayed. The header displays the function that is currently being executed (here: *Interrupt operation).*





In the first dialog *Enter quantities,* the user can enter the produced yield or scrap quantities. Subject to the active input field, the virtual keyboard is shown or hidden automatically.

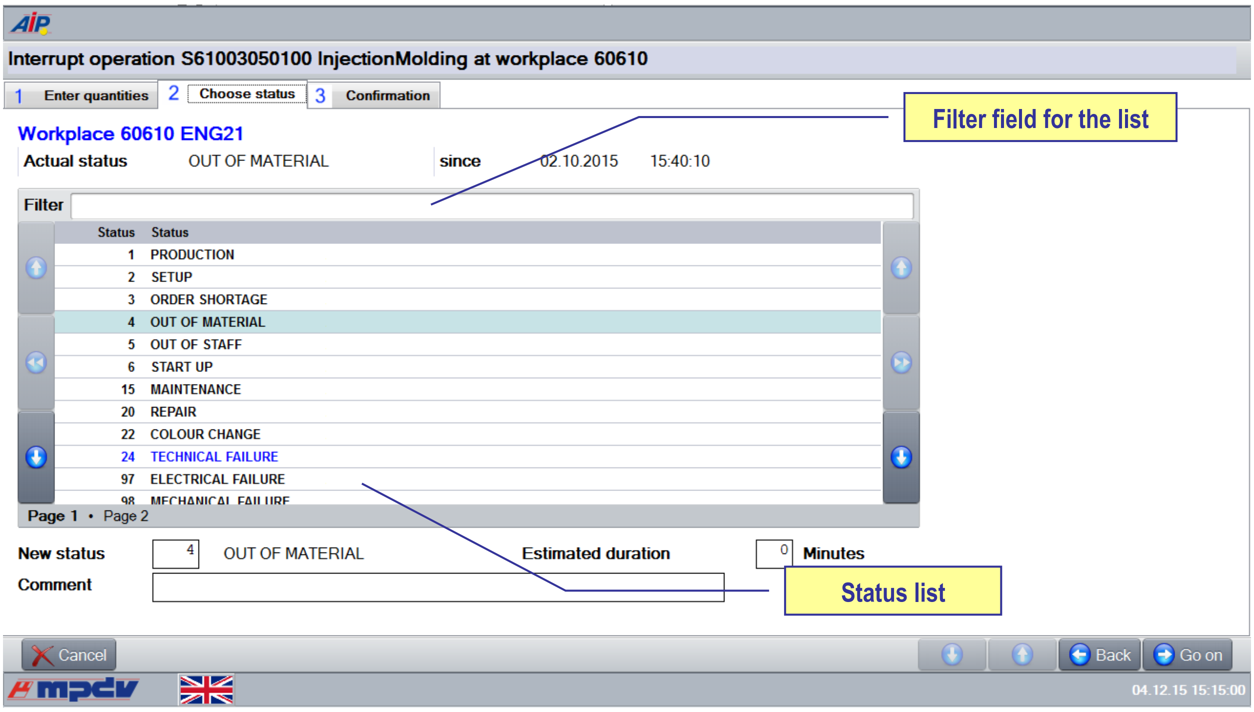
Quantities can be entered using the virtual or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). When the user has entered all values in the first view, the next view (sub-dialog) can be opened by clicking ***Next***.

The ***Cancel*** button is displayed in all sub-dialogs. Click this button to cancel/close the entire process at any time.

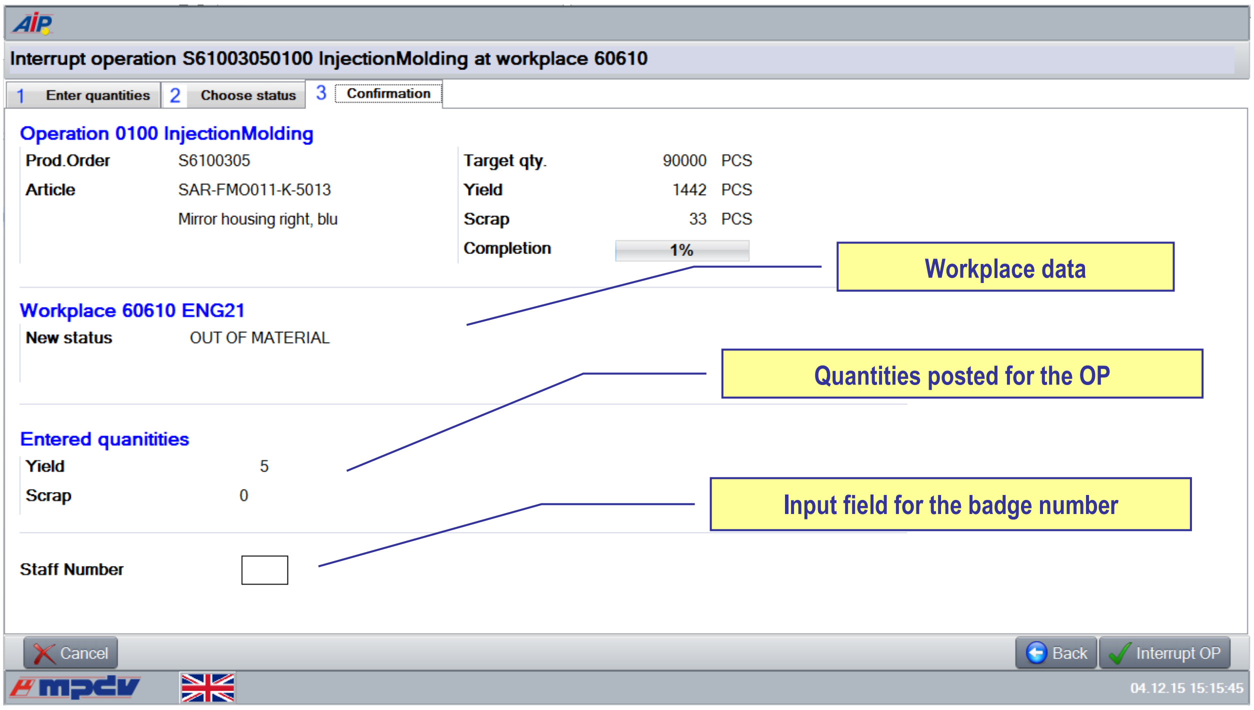
To open the next view (***Select status*** in the example), click the ***Next*** button or another tab (in our example: ***Select status*** or ***Confirmation).*** Please note in this context, that no view can be skipped when they are navigated upwards (view 1 → view 2 → view 3). This means: When you are in the first view *(enter quantities)* and you click the third view *(confirmation),* the second view *(select status)* will be displayed first.

Vice versa, when navigating downwards (e.g. from the *confirm* view to the *enter quantities* view), each view can directly be opened by clicking the required tab. In this case, views can actually be skipped. Using the ***Back*** button, views are opened one after the other (upwards).

As long as the dialog has not been confirmed, the data entered can be changed at any time by scrolling back and forth.



In the second view *Select status*, you select the workplace status that is set, when the operation has been interrupted. You can select the status from the status list displayed. This list can be restricted using the *Filter* field. Once the required values have been entered, the next view/sub-dialog can be opened by clicking ***Next*** (in our example it is the last view).



The sub-dialog *Confirmation* shows a summary of all values entered in the dialog. If the user agrees with the entered data, the ***Interrupt operation*** dialog can be confirmed, once the badge number has been entered. Then the dialog is sent to the server and posted.

If an input field of the dialog is not completed properly (e.g. a mandatory field is empty), the field is highlighted in red in the respective view and gets the focus. The user can then directly correct the value.

If a workflow dialog is opened, you can click the ESC key to directly exit the dialog. This exit is also possible, if the virtual keyboard is displayed. As a consequence, you cannot use the ESC key to close the virtual keyboard.

3 Main View with Tiles

With the AIP2, the user can switch between the tile design optimized for touchscreens and the list format. By default, the tile layout is shown, which is described in the sections that follow.

To ensure proper processing and posting, terminals with "MDE" operation mode must <u>not</u> be switched off during times without shift.

3.1 Main view – header and footer

Header

The *AIP* logo is displayed on the top left of the screen, which may be replaced with a customer logo after configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Main views

You can assign a maximum of 16 workplaces or machines to the *AIP2* terminal. The different workplaces are listed in the order that they were assigned to the terminal on the client. .

In the main view of the *AIP2,* you can use the button "< Overview“ to switch to the icon view of workplaces. In the terminal configuration of the client you can specify whether you want to use the icon view. The sections that follow describe the main view and the icon view.

Footer

The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where you can start further administration functions. This dialog closes automatically after approx. 5 seconds.

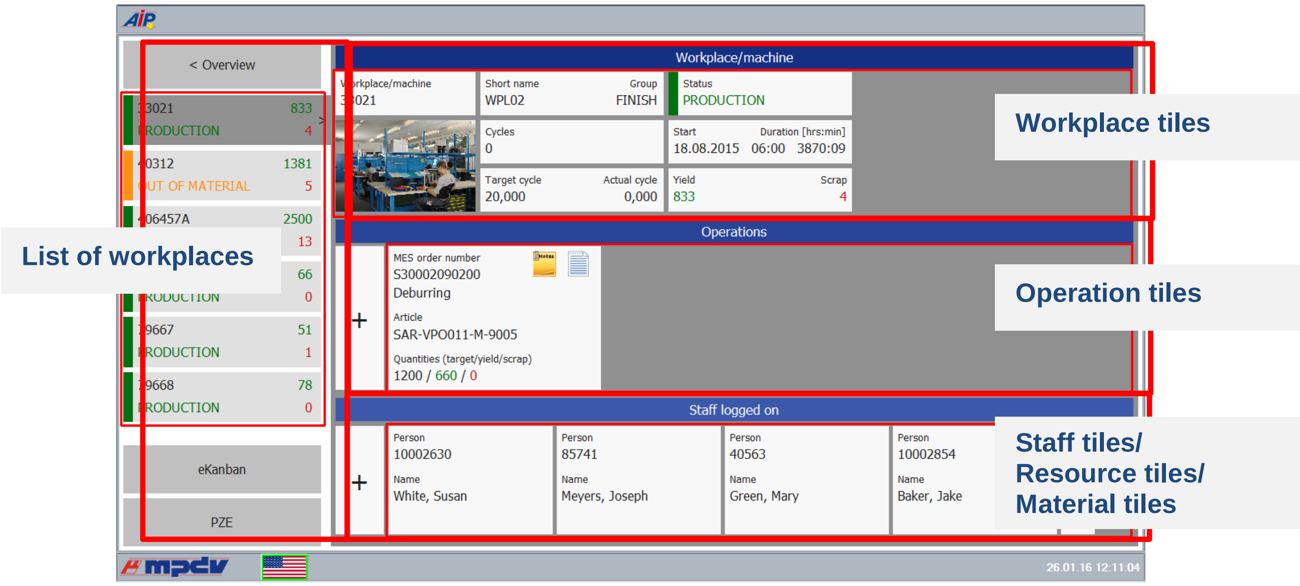
In the middle, further information is displayed: the current terminal status, AIP2 version number, date of the build, IP address of the server and the terminal number.

The current date and time are displayed to the right.

AIP2 statuses

	Network connection has been established. The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But you can still record certain postings. These postings are transferred to the server, once data connection has been re-established.
	Data is being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending saved data records to the server.
	DEMO mode The terminal is in DEMO mode, i.e. server communication is disabled.

3.2 Main view with "tiles"



Please note: The actual display can be different to the above illustration.

Subject to the configurations made, the main view with tiles consists of two or three rows of tiles. While the first two rows of tiles (workplace and operation tiles) are always displayed, it is up to the user whether or not the third row of tiles is shown (optional display). In the configuration of workplaces you can configure for each workplace separately if you want to show the third row of tiles.

Additionally, there is the list of workplaces to the left. Here, you can select the workplaces for which details are displayed on the right-hand side.

List of workplaces

The workplace list shows all workplaces/machines assigned to the terminal. If many workplaces are assigned, swipe to get to the workplaces displayed further down.

This information is shown for the workplaces:

Machine/workplace number

Shows the machine and/or workplace number.

Status

The status is displayed for each machine in color on the left-hand side and also the status text is colored. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Quantities

On the right-hand side, the first figure shows the produced yield in green and the red figure shows the produced scrap.

If you have enabled the *Compensate manual quantities* option (e.g. set off scrap against yield) and the machine list also shows shift-related quantities, they will not be updated immediately. The application only updates the quantities, once the lists have been reloaded.

Unit for yield and scrap

If no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on, the primary quantity unit of the operation is displayed.

Workplace tiles

The workplace tiles provide the following details:

Workplace/machine

No workplace or machine number is displayed.

Short name / group

The short name of the workplace or machine and the machine group are displayed.

Status

The status is displayed in color and as status text on the left-hand side. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Machine image

Shows the picture of the machine stored in the configuration of workplaces.

Clocks

Shows the recorded machine cycles of the current shift.

Start / Duration [hrs:min]

Point in time since the status has been available and the resulting duration at the current point in time.

With BDE workplaces[1], this point in time refers to the last manual status change. With MDE workplaces, it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was last changed manually or to the time of the last shift change.

Target/actual cycle

Current target and actual cycle of the workplace.

The largest target cycle of all operations logged on to the workplace is shown. The largest target cycle is transferred to the MDE for monitoring.

If the target cycle is smaller than the minimum cycle time, the target cycle is still shown.

If an operation is logged off or interrupted, the largest target cycle of the remaining operations is identified and displayed. After logoff or interruption of the last operation at the workplace, the last target cycle set is still displayed.

If no operation is logged on, the target cycle specified in the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that last applied.

Yield / Scrap

Yield and scrap quantities of the current shift produced at the machine/workplace.

KPIs: OEE, utilization efficiency, scrap ratio

This function is only available if you enable the extension aipkpi.

Shows the KPIs OEE, utilization efficiency and scrap ratio. The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation“). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached. If you have not defined limit values, the application shows the KPI in gray, otherwise in the color you defined for exceeding/not reaching limit values. For further information on the configuration, refer to the document MDE_KPI_Configuration.pdf.

[1] An MDE workplace is a workplace that is assigned to a terminal, which runs in the “MDE” operation mode. Otherwise, it is a BDE workplace.

The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

Linked functions

If you click the workplace status, the dialog for changing the status opens. If you click one of the other tiles, the dialog opens where you can start the functions available for the selected workplace.

The buttons displayed depend on the selected workplace. The button *Lock production status*, for example, is only available for MDE machines.

List of operations logged on

The middle area on the right-hand side shows the logged on operations as tiles. The following data is shown:

MES order number

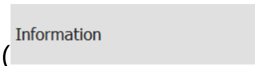
Order number and operation number of the operation logged on. The combination of these two numbers is the *MES order number.*

Article

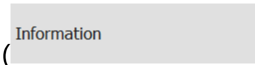
Article defined for the operation.

Quantities (target / yield / scrap)

Shows the target quantity defined for the operation, the produced yield and the scrap. The yield and scrap quantities integrate the counter readings of the available machine connections.

This icon is displayed if at least one note has been recorded for this operation in the graphic planning board of the client that must be shown on the terminal. To display the note(s), click the icon or click the operation and select the *Information* button (

).

This icon is displayed if at least one long text is stored for this operation. Click this icon to display the long texts or click the operation and select the button *Information* (

).

If you click an operation, a screen opens that shows the workplace/operation data. Via this screen, you can also select the operation-related functions.

Operation tiles

In addition to the fields already described, the operation tiles also show the following data:

Comments

This tile shows the user fields 53 and 54 (alphanumeric, 20 characters) of the operation. To edit these fields, you must store a respective user field key for the operation, which includes these two fields.

Completion in %

The bar shows the proportion of "yield", which has been produced until now, compared to the "target quantity".

Since logon (target / yield / deviation)

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time when no production lock has been set for the machine). If the terminal program has been restarted after the OP logon, no value can be calculated.
Calculation:
Target Since Logon = Net Running Time[sec] * Partitioning/Target Cycle[sec/stroke]
Net Running Time: Time since logon while the production lock has not been set. This calculation does not integrate the breaks specified in the shift model or the status times posted to RPA 12 (resource performance account).

Deviation (in percent) between the calculated target quantity since logon and the quantity which has actually been produced "since logon".
Calculation: Deviation[%] = 100% * (Yield Since Logon - Target Quantity Since Logon) / Target Quantity Since Logon.

As of CTAIP 8.2.1.32:

Up to now, the target quantity since OP logon was only calculated if the workplace/machine was assigned to a terminal with operation mode "MDE processing". As of CTAIP 8.2.1.32, the target quantity since OP logon is also calculated if the terminal is configured with operation mode "BDE processing".

If the terminal has been restarted after an operation logon, the target quantity cannot be calculated correctly. To improve transparency, an "*" (asterisk) is shown behind the target quantity (since OP logon) in this case. The asterisk indicates that the target quantity now displayed no longer refers to the time of the operation logon, but to the time of the terminal restart.

While the workplace/machine status 999 is displayed, the target quantity since OP logon is "---". If the status 999 is again changed within a "free shift", the target quantity since OP logon is calculated using the point in time of the OP logon, of the shift start or of the terminal start.

To disable the calculation of the target quantity since OP logon on the terminal, you can use the configuration CalcTargetYieldSinceLogon=0 in the hytnrcfg.ini. The quantity is then displayed using "---".

With workplaces/machines that are configured as "Machining centers", the calculation of the target quantity since OP logon is generally disabled because this calculation contradicts the principles of the machining center.

Planned duration

The field *Planned duration* displays the target processing time of the operation in format [h:mm].

Partitioning

Calculate the displayed partitioning as follows:

TLG_M	Partitioning of the workplace/machine (TLG in mnr.lst)
DIV_M	Pulse factor of workplace/machine (IMPFAKT in mnr.lst)
TLG_{AG}	Partitioning of the operation (TLG in anr.lst)

The application shows the calculated partitioning without decimal places, provided it is an integer value.

In case the partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Displaying the "3rd list"

The third list is optional. You can configure the third list in the configuration of workplaces. The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- List of materials/input batches (MPL/TRT) logged on to the currently selected workplace
- List of output batches produced in the currently selected operation (MPL/TRT)

In case you have enabled several lists, you can switch between these lists in the header line that is located above the third list. Activated lists can be selected one after the other.

Maintenance status

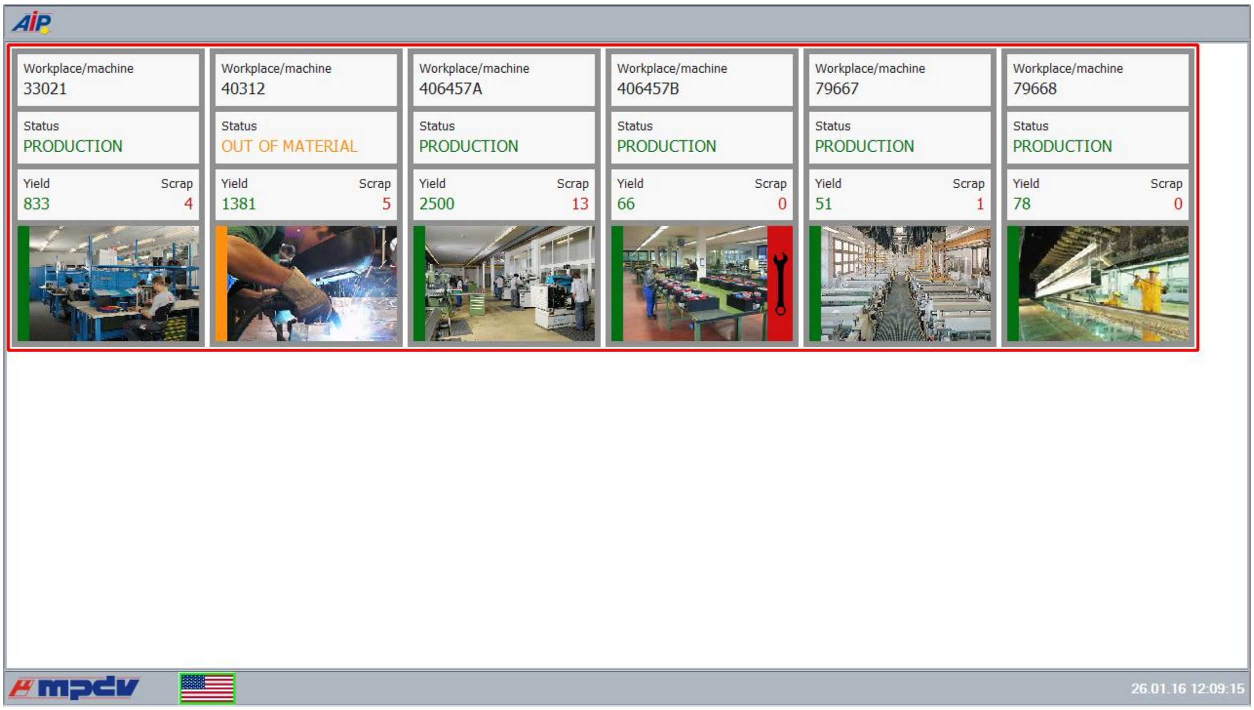
If you have purchased the license for the maintenance calendar, the maintenance status is displayed using a yellow or a red field showing a wrench. The color displayed depends on the required maintenance activity.

Calling functions

The functions available are assigned to the relevant objects. Example: The functions *Log person off* and *Log all staff off* are displayed if you click on a person logged on.

3.3 Icon view of workplaces

You can enable this view in the configuration of terminals via the client. Then open this view by clicking the button "< Overview“ in the main view. This view shows workplaces in a clear structure and with an image. It shows important information on the single workstations:



Please note: The actual display can be different to the above illustration.

A colored bar to the left of the image indicates the current status of workplaces/machines:

- green: production
- yellow: assigned status
- red: not assigned

Each tile includes the workplace/machine number, the status (text) of the workplace, the yield and scrap quantity and the image of the workplace.

A colored background with a caliper and/or wrench to the right of the image indicates if an inspection or maintenance is due for the workstation.

If you enable the extension aipkpi, the application also shows the KPIs OEE, utilization efficiency and scrap ratio.

The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation“). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.

The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

If you click on a tile, the previously described main view is displayed and the workplace is automatically selected. From there, you can perform the postings for the selected workplace.

Use the option "< Overview" to exit the main view and to return to the icon view.

As part of the advanced configuration options, you can customize the layout of display lists, the displayed data fields and functions. For technical reasons, however, you cannot change the sort sequence of display lists in the main view of the terminal.

As of AIP 8.2.2.28, you can automatically change from the main view with tiles to the icon view of workplaces after a configured time. The configuration `AUTOMATIC-CHANGE-TO-START-DISPLAY` is described in the document AIP2_Configuration_hytnrcfg.pdf.

4 Basic Screen as List View

The new tile design can be disabled for the AIP2 terminal. This chapter describes the basic screen with disabled tiles.

In general, the AIP2 has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.

To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

4.1 Basic screens – header and footer

Header

The *AIP* logo is displayed top left of the screen, which may be replaced with a customer logo after corresponding configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Basic screens

A maximum of 16 workplaces or machines can be assigned to the *AIP2* terminal. The single workplaces can be found within the list area in the order assigned to the terminal via the client. .

As regards the basic screen of the *AIP2* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured via the configuration of terminals in the client. The single basic screens are described in the sections that follow.

Footer

The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

Further information is displayed in the center : the current terminal status, AIP2 version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

Terminal status

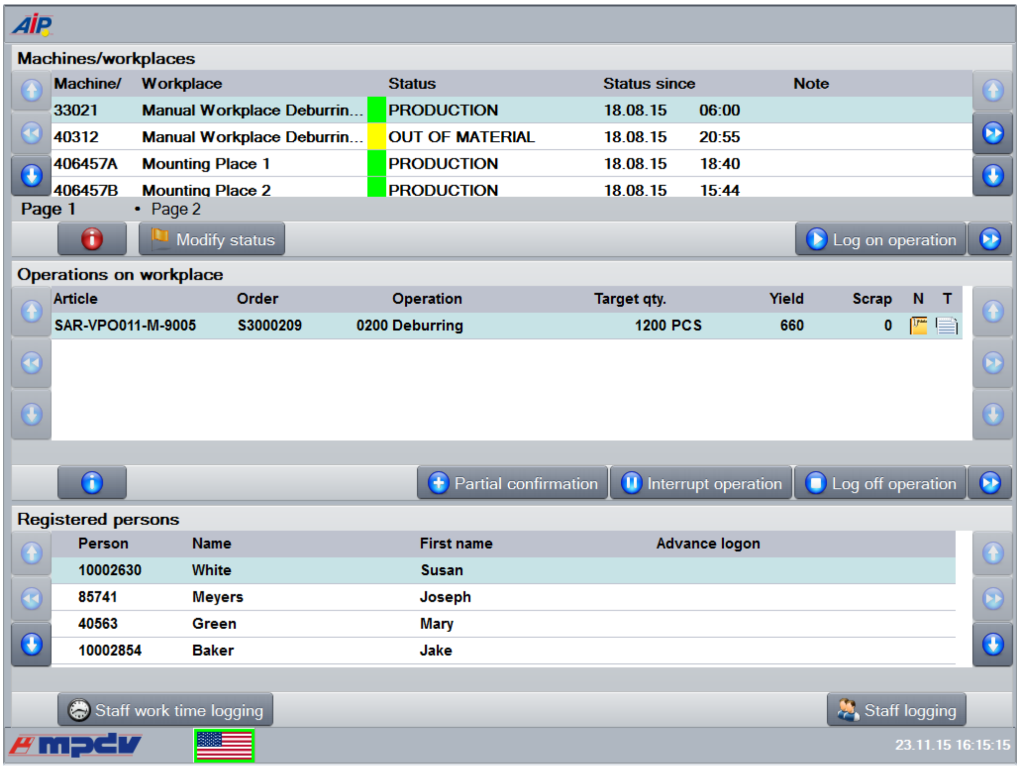
	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

4.2 Basic screen “tabular view“

1st list
Workplaces assigned to the terminal

2nd list
List of registered operations

3rd list (optional)
e.g. list of registered staff



Subject to the configurations made, the tabular basic screen consists of two or three tables. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional).

“Machines/workplaces" table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as a description are displayed.

Status

The status is highlighted in color and the status text is shown. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

Status since

Point in time since the status is available.

For ADE workplaces[2] the point in time refers to the last manual status change. For MDE workplaces it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was changed manually most recently or to the time of the last shift change.

Please note:

It is indicated here if the “lock production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.

By way of “customizing” services it is possible to adapt the layout of the display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic screen of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately. Quantities are only updated once the lists have been reloaded.

"Operations at workplace" table

The second table shows the operations currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number.*

Target quantity

Target quantity defined for the operation.

[2] We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an BDE workplace.

Yield

Yield already produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity already produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note visible on the terminal has been recorded for this operation in the graphic planning board of the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button ).

Below the second list there is a row that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured. Information displayed in this list depends on the workplace configuration.

The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow switching between these lists.

Please note

The staff logged on displayed in the third list is identical to the list displayed in the dialog “F5 staff logged on…”. Selecting a person in the third list does *not* affect the selection of the operation in the list of OPs running at the workplace. Therefore, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic screen

A toolbar, which may be customized, is assigned to each list included the basic screen. This makes the purpose of a function clear to the user. The “partial upload/confirmation” function can be found below the list of registered operations.

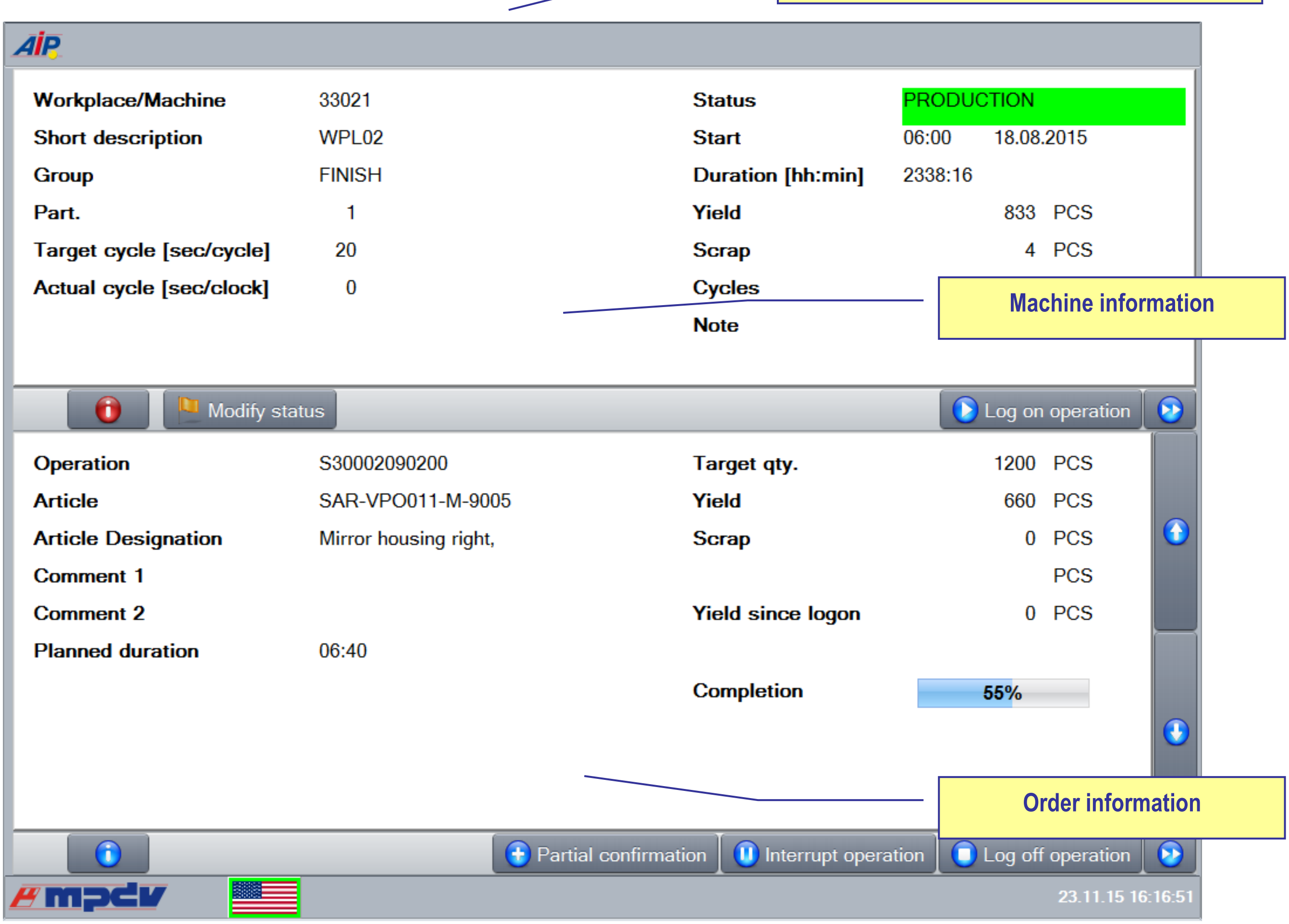
In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. *change partitioning*) can be opened by clicking the corresponding button.

Please note

The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

4.3 Basic screen "machine overview"

If the “change view” button is clicked in the basic screen, the view changes to the following presentation:



This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation consists of three sections:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines and shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. If several orders/OPs are logged on to the workplace, then extra arrow buttons are displayed. It is possible to switch between individual orders/OPs using these arrow buttons.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \cdots\right]$$

TLG_M	Partitioning of the machine (TLG in mnr.lst)
DIV_M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG_{OPi}	Partitioning of the individual operation (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview of the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) of the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time while the production status was not locked for the machine). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:
TargetSinceLogon = NetRunningTime[sec] * Partitioning/TargetCycle[sec/stroke]
NetRunningTime: Time since logon while the production lock has not been set. This calculation does not take into account any breaks defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

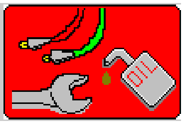
Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (YieldSinceLogon - TargetSinceLogon) / TargetSinceLogon

Completion

The bar represents the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced with a picture showing a yellow or red oilcan. It all depends on the required maintenance activity:



4.4 “Machines as icons” basic display

This view can be configured as the default display in the configuration of terminals at the client It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green:	Production
- yellow:	Assigned status
- red:	Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched , the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon view of machines”.

5 BDE and MDE Functions

This document describes the different *AIP* functions used for the Shop Floor Data and Machine Data Collection (BDE and MDE).

5.1 Logging on an operation

When you log on an operation, the following methods are available:

Separate logon of orders and staff	Operations and staff have to be logged on/off separately. You have to enter the staff badge number with the function **Log on operation**. This is a validation check only. The person must be logged on separately using the function **Log on person.**
Log person on with order	The OP and the person are logged on in one posting process. You only use the function **Log on person** if further persons are logged on to this operation.

Make this setting in the terminal configuration on the client (*Log user on with order* option).

Posting process

Select a workplace before you log on an operation. When you then call the dialog, the workplace field is already populated.

Calling the function *Log on operation*

Click the button *Log operation on.*

When the function is called, the user is guided through the input dialog.

Choose operation

Manual entry via keyboard

or

Selection via sequencing list (see the "Notes on the sequencing list")

or

Scan of bar code

With manual entry or bar code scan, the operation is not automatically searched and positioned in the sequencing list.

Choose status

Enter or select the status number that must be set when the operation is logged on.

Staff badge number

The input field is also displayed if you have selected the “separated“ option for the "logon" in the terminal configuration (client). The person is not logged on with the order but is only used for validation. For this setting, go to the terminal configuration*: authorizations on the terminal > log OP on*.

Confirmation *Log on operation*

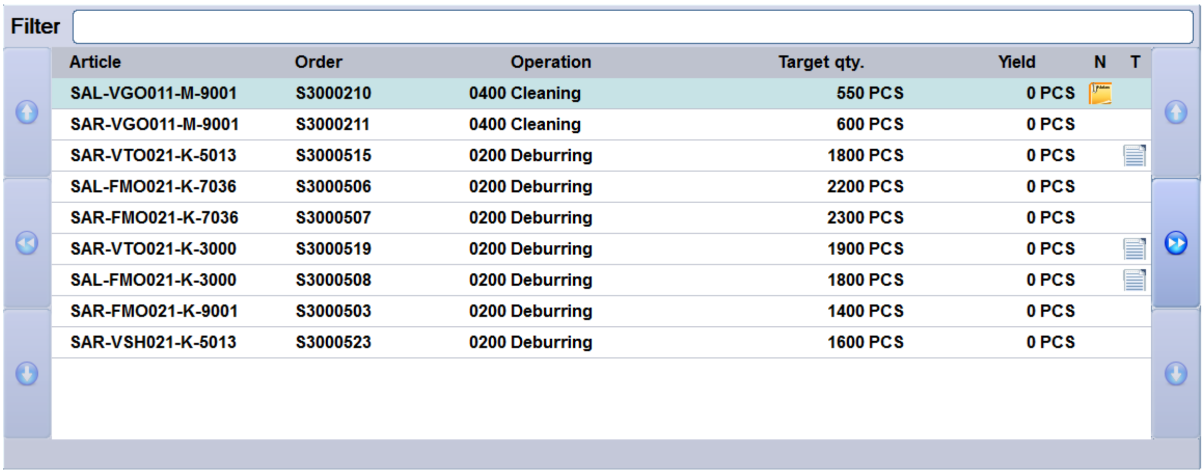
You use the button ***Log on operation*** to log on a new operation. With the OP logon, the run times are booked to the different time accounts and the quantities are posted for the OP.

When the operation is logged on, the *Workplace* field in the backlog of orders is filled with the workplace where the operation is logged on. If the operation has been planned for another workplace, the planned workplace is then overwritten. As a result, the OP is implicitly re-planned. This re-planning does not involve any further actions (e.g. update of the template, rescheduling).

Notes on the sequencing list

You can limit the number of entries in the sequencing list for the specific workplace. The list should not get too long because a long list has a negative effect on response time behavior, operability and search time (recommendation: not more than 50 entries, less with remote connections).

The same is valid for the functions *Interrupt operation, Finish operation* and *Partial confirmation*. Here, running operations are displayed.



Filter

Article	Order	Operation	Target qty.	Yield	N	T
SAL-VGO011-M-9001	S3000210	0400 Cleaning	550 PCS	0 PCS		
SAR-VGO011-M-9001	S3000211	0400 Cleaning	600 PCS	0 PCS		
SAR-VTO021-K-5013	S3000515	0200 Deburring	1800 PCS	0 PCS		
SAL-FMO021-K-7036	S3000506	0200 Deburring	2200 PCS	0 PCS		
SAR-FMO021-K-7036	S3000507	0200 Deburring	2300 PCS	0 PCS		
SAR-VTO021-K-3000	S3000519	0200 Deburring	1900 PCS	0 PCS		
SAL-FMO021-K-3000	S3000508	0200 Deburring	1800 PCS	0 PCS		
SAR-FMO021-K-9001	S3000503	0200 Deburring	1400 PCS	0 PCS		
SAR-VSH021-K-5013	S3000523	0200 Deburring	1600 PCS	0 PCS		

If an operation is already logged on to the workplace, then the system enters this workplace in the input field *Operation.* If you do not want this and if the system should preset the first operation of the sequencing list in this field, then you must make the following configuration:

- Call the application Dynamic dialog configuration - Fields.
- Request data for the dialog "WF_AGL". If it is not the logon dialog A_AN or A_P_AN, then identify the dialog.
- Select the field "ANR".
- Click the *Edit* button (note: this is only possible with terminal-specific or terminal group-specific dialogs. It is not possible with AIPDEF 0 dialogs).
- Enter the value "SETVALUE" (without quotation marks) in one of the field attributes 1 to 8.
- Leave the field *Default* empty.
- Save your changes.
- Change to the application Dynamic dialog configuration.
- Restart the dialogs and then restart the terminals.

Displaying notes and texts

Via configuration in the application Order types, you can use the option *Show OP info when logging on OP* to specify for a specific order type that after a successful order logon on the AIP either

- the notes on the operation with an active option *Visible on the terminal*

or

- the long texts of operations

are automatically displayed. The “OP info” dialog is then opened with the respective active page.

Note: To show the information, the AIP must be connected online to the server.

5.2 Logging an operation off

You use the button *Log off operation* on operation level to log off an operation. The posting of run times and quantities is then finished for the OP. Once logged off, you cannot log on the OP again.

Posting process

Select the workplace and the operation that must be logged off in the main view. When the dialog is called, these fields are then preassigned and cannot be changed.

Calling the function *Log operation off*

Click the *Log operation off* button.

When the function is called, the user is guided through the input dialog.

Yield

Enter the yield quantity that must be posted.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). Enter the produced scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Deviation reason

You can activate an overdelivery or underdelivery check for operations or persons. When the operation is logged off, the system then checks if the quantity recorded has exceeded the overdelivery limit or falls short of the underdelivery limit. If the check has a positive result, the quantities posted are rejected with the error message "Overdelivery" or "Underdelivery".

If only a warning is activated for the check, the user can force the system to accept the overdelivery or underdelivery by entering a deviation reason.

Status

Enter or select the number of the workplace/machine status that is set after the operation is logged off.

Estimated duration

Optionally enter the estimated duration of the selected workplace/machine status in minutes.

Note: Only make an entry here if the status is a downtime status (status <> production).

Comment

Enter an optional comment for the status in this field that is also displayed in the machine history.

Staff badge number

The person entered must be authorized to log off the operation. The setting is made in the HR master data. Go to: Shop floor data > BDE authorizations > Lop OP off.

Confirming *Log off operation*

The operation is logged off, once the dialog has been confirmed. You cannot log the operation on again.

5.3 Interrupting an operation

Click the button *Interrupt operation* to call this function. You use this function to stop collecting times and quantities for an order. The reasons for the interruption can be a quantity upload, a shift change or an interruption of the production for technical reasons.

The process of interrupting an operation and the layout of the input dialog are identical to the ones of the operation logoff. The difference is that you can log on an interrupted operation at any time.

5.4 Uploading a part quantity for an operation

You can use this function to upload a part quantity for the order without interrupting or finishing the running OP. The quantities are booked for the relevant OP and the person that makes the posting. Click the button *Confirm partially* to call this function.

Posting process

Select the workplace and the operation from the list. When you then call the dialog, the system preassigns these fields and the values cannot be changed.

Calling the function *Confirm partially*

Click the button *Confirm partially.*

When the function is called, the user is guided through the input dialog.

Yield

Enter the yield quantity that you want to post.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). You can enter scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Deviation reason

You can activate an overdelivery or underdelivery check for operations or persons. The system then checks if the overdelivery limit is exceeded with the quantity posted. If the check has a positive result, the quantity posted is rejected with the error message "Overdelivery".

If only a warning is activated for the check, the user can force the system to accept the overdelivery by entering a deviation reason.

Staff badge number

The person entered must be authorized to upload part quantities for the operation.

5.5 Log person on

Persons are logged on to or off from a workplace. The logging of persons is therefore made on the workplace level.

You can only log on a person, if an operation has been logged on to the workplace.

You can only use this function to log on persons with single workplaces or production orders.

With group workplaces or overhead cost operations, the persons are logged on via the combined logon of order and persons when the operation is logged on. It must be guaranteed here that a staff badge number can be entered during the posting process.

Posting process

Select the workplace where you want to log the person on or off and click the button *Staff logging*. The system displays a list of the *Registered persons* that are already logged on to this workplace. Close this view by clicking *Close information*.

Calling the function *Log person on*

Click the button *Log person on.* A dialog including only one dialog step opens.

Staff badge number

To identify the person, also enter the staff badge number here.

Other notes

Via customization, you can optionally enter an operator position/function or a premium indicator. You specify in the course of the customization how the value is entered: only a field with a selection list or a separate workflow.

You can configure that an advance logon for the next shift is possible within a configurable time before the start of the next shift. Configure this time in the terminal configuration (tab: *MF functions > Waiting period for advance logon of staff*). But if the last operation is logged off before the end of the current shift, all advance logons are deleted. And you cannot log off a person that is logged on in advance.

5.6 Log person off

Persons are logged on to or off from a workplace. The logging of persons is therefore made on the workplace level. You can only log off a person that has been logged on to the workplace.

You can only use the function to log off persons with single workplaces or production orders. With group workplaces or overhead cost operations, the persons are logged off via the combined logoff of order and persons when the operation is interrupted or logged off.

Posting process

Select the workplace where you want to log the person on or off and click the button *Staff logging*. The system displays a list of the *Registered persons* that are already logged on to this workplace. Close this view by clicking *Close information*.

Calling the function *Log person off*

Click the button *Log person off.* A dialog including only one dialog step opens.

Yield

Enter the yield quantity that you want to post.

Scrap (input fields and display)

The scrap reasons are already displayed in the dialog (see also AIP operation). You can enter scrap quantities for the relevant scrap reason. All scrap quantities entered are totaled and displayed in the general display field.

Staff badge number

To identify the person, also enter the staff badge number here.

5.7 Log off everyone

You can use this function to log off all persons in one posting that are logged on to a machine.

You assign the authorization to use this function in the HR master data. Activate the following option in tab *Shop floor data* > *BDE authorizations* > *Log all staff off.*

Posting process

The posting process is in general identical to the dialog *Log person off.* The only difference is that you cannot record quantities when using the dialog *Log off everyone*.

5.8 Change workplace/machine status

You can use this function to assign a new status to a workplace/machine. You might need this function during setup of the workplace or in case of a malfunction, for example.

A status should be entered, when the AIP automatically identifies a malfunction and the status changes to “not assigned” (also see "Monitoring of operating signals and cycle time").

You can configure that only persons can perform a status change that are already logged on. Enable the following option in the HR master: tab *Shop floor data > BDE authorizations > Change only if person is logged on.*

Posting process

Select the workplace where you want to change the status and click the button *Change status*.

Status

Enter or select the status that you want to set for the workplace.

With manual entry or bar code scan, the status is not automatically searched and positioned in the selection list.

Estimated duration

You can enter an estimated time in minutes that the new status will probably take. This field should only be filled for downtime statuses.

Comment

You can enter a comment for the status. This comment can be displayed in the machine history.

If the status lasts on and shifts change, then the comment is only assigned to the MDE log record of the shift before shift change.

Staff badge number

Enter the badge number of the person that changes the status. The number is required for the validation check. The person must be authorized to change the status.

Confirmation of the dialog

Once the dialog has been posted successfully, the new status is activated for the machine.

Other notes

It is possible to change from one status to another, unless the automatic status monitoring has been activated for the workplace/machine (workplace/resource configuration > *tab: MDE configuration > automatic monitoring: cyclic or operating signal*) and the workplace is in status “production”.

If you do not want to show a machine status in the status list (any more), then you must disable the option *Status manually at terminal* in the machine status assignment.

Hierarchical statuses

You can build a status hierarchy as of HYDRA-MDE 7.2. In the status assignment, you store the status number of the direct superordinate status for a lower-level status. All statuses that you cannot assign on the terminal and that are only used to show the hierarchy are called “hierarchy level”.

Hierarchy levels are displayed in blue font in the status list.

If you select a hierarchy level and double-click/touch the button, the list of “lower-level” statuses opens. To get to the next higher hierarchy level, click/touch .

You can only select a status in the status list that can be assigned manually. The following error message is shown, if a machine status change on a hierarchy level cannot be assigned.

5.9 Lock/unlock production status

The button *Lock production status* or *Unlock production status* is only active for a workplace if the current status is not "production" and if one of the following conditions is fulfilled:

- The PCC receiving the machine data of the workplace runs in embedded mode and the workplace is assigned to a terminal configured with "MDE operation".
- The PCC receiving the machine data of the workplace runs in stand-alone mode and the INI configuration MDE-NOTIFICATION is activated for the workplace.

If the function *Lock production status* is active, the terminal cannot automatically change to status "production" when the terminal identifies signals (pulses or operating signal).

If the production status is locked, this means:

- With an active lock, the automatic change to status "production" is not possible and the status currently set is kept despite machine pulses (e.g. "setup").
- If cycle signals are recorded during the lock via counter inputs configured as "yield", then the relevant quantities are booked according to the Configuration of *Posting during prod. lock*:
    - as yield,
    - as scrap
    - or not at all.
- Click the button *Unlock production status* to remove the lock.

The production status can also be locked/unlocked when a specified workplace/machine status is set. You can configure the behavior in the configuration Status assignment for each status that is not "production" (go to: *Master data* > *Workplaces/machines* > *Status assignment*).

Logging of the production lock

The manual lock or unlock of the production status is logged as event on the server and can be shown in the Machine history. Note: the logging of the event is only performed if the status is manually locked or unlocked.

But if the production status is locked or unlocked via status change, then this is not documented explicitly as an event and is therefore not shown in the machine history.

If the production status is locked for a machine and the terminal software is restarted, then the production lock is automatically removed after this restart. The changed production lock is not logged.

Authorization check when setting the production lock manually

You can configure that only if a relevant authorization is available, the person is allowed to manually (explicitly) set or remove the production lock.

To do so, activate the dynamic dialog M_PSPERRE. If this dialog is activated, you must enter the staff badge number.

If the dialog is activated, the dialog is displayed when the operator clicks the button *Lock production status* or *Unlock production status*. Once the badge number has been entered, the system checks whether this person is authorized to lock/unlock the production status. The system checks, if the option *Change of production lock* in the HR master data (tab *Shop floor data*) is enabled.

If the terminal is OFFLINE, the terminal configuration specifies the behavior (option *Checking required*).

5.10 Change target cycle

The target cycle is the default value that is checked in case of a machine monitoring based on cycle time. You can use this dialog to change the target cycle for the machine.

Posting process

Select the workplace where you want to change the target cycle. Click the button *Change target cycle.*

New target cycle

Specify the new target cycle in seconds/cycle

Staff badge number

The entry of the staff badge number is optional. If configured accordingly, a validation check is performed for the number. (*HR master data > tab: Shop floor data > Change of cycle/partitioning*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new target cycle is activated for the machine.

5.11 Change partitioning

The partitioning (also called cavity) specifies the number of parts produced per machine cycle (clock). You can use this dialog to change the partitioning. The automatic collection of quantities is then based on this partitioning.

Posting process

Select the workplace where you want to change the partitioning. Then click the button *Change partitioning.*

Partitioning

Enter the new partitioning.

Staff badge number

You can optionally enter the staff badge number. If configured in the HR master data, a validation check is performed for this number (tab: *Shop floor data > BDE authorizations > Change cycle/partitioning*).

Confirmation of the dialog

Once the dialog has been posted successfully, the new partitioning is used for all running operations at the machine.

5.12 Change target quantity

Use this dialog to change the target quantity based on operations (primary quantity unit).

Posting process

Select the operation where you want to change the target quantity and click the button *Change target quantity.*

New target quantity

Enter the new target quantity that you want to store for the operation.

Staff badge number

You can optionally enter the staff badge number. If configured in the HR master data, a validation check is performed for this number (tab: *Shop floor data* > *BDE authorizations* > *Change target quantity*).

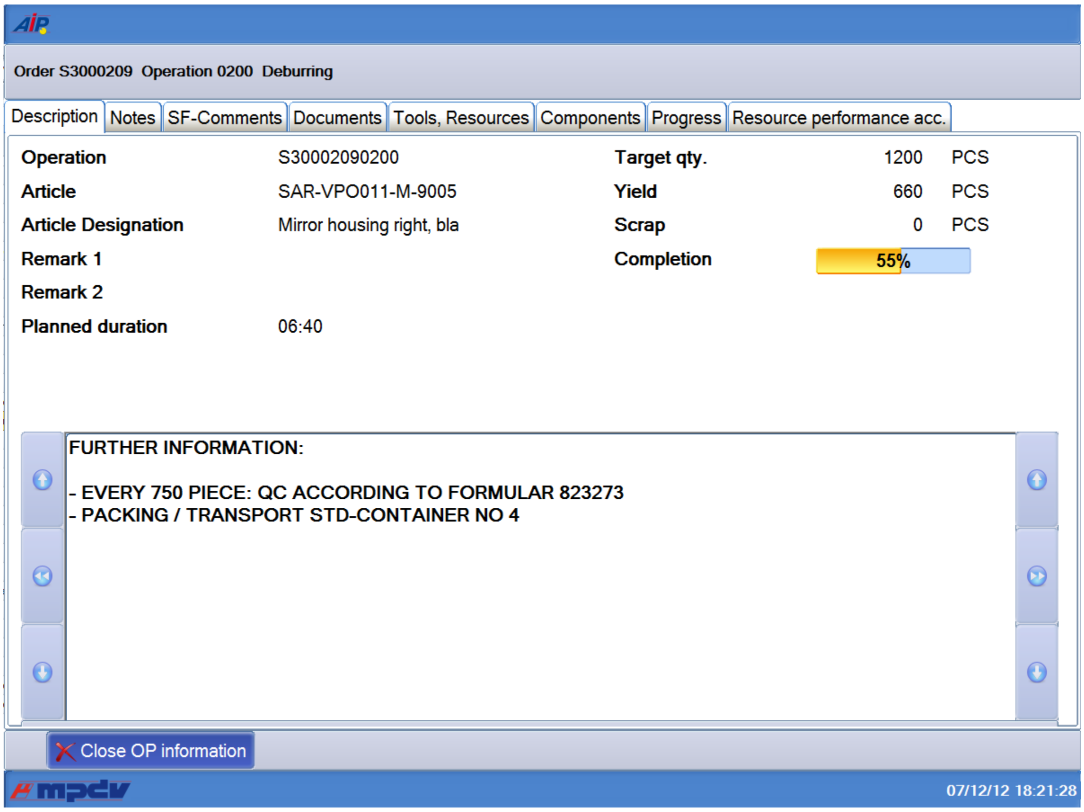
Confirmation of the dialog

Once the dialog has been posted successfully, the new target quantity is stored for the operation.

5.13 Information on operations (OP info)

General information

Use the button  on operation level to call the OP info dialog[3]. Select the required operation. A dialog opens. The dialog includes several pages that are organized in tabs. The information is only requested from the database when you call the relevant page.



The dialog includes the following tabs:

[3] This function is also available at other places, e.g. in the *Log operation on* dialog.

Description

Current information on the current operation is displayed.

- Operation (MES order number)
- Article
- Article designation
- Remark 1
- Remark 2
- Planned duration
- Target quantity (of the operation)
- Yield
- Scrap
- Completion (in %)

At the bottom, the long text assigned to the operation is displayed.

Notes

The tab *Notes* displays the notes entered on the client (usually via the *Graphic planning board*) if the option *Display on terminal* is enabled.

The table shows all notes of the operation that are configured to be displayed on the terminal. The list is sorted by the editing date. The most recent note is on top and is displayed when you call the view.

If you click/touch a note of the list, the complete text is displayed in the field below.

5.13.1 SF comments

The tag *SF comments* displays the comments recorded during the Shop Floor Data Collection (BDE) or you can record new comments.

The comments recorded are displayed in the *Order information* dialog on the client or can be forwarded via escalation management, e.g. by e-mail.

If an SF comment is recorded for an operation that has been merged on the MOC, then the SF comment is only relevant for this merged operation. The SF comment is not transferred to the single operations.

You cannot record SF comments for the single operations because the single operations are not displayed on the AIP.

With split operations, the SF comment is only stored for the split operation where the SF comment has been recorded. The SF comment is not transferred to the split master.

5.13.2 Documents

You can display documents, graphics or other files on the AIP, which are listed in the table in tab *Documents*.

Select an entry in the list and touch/click the button *Open document*. The file is downloaded to the AIP and displayed using an internal viewer or an external application (according to the file extension).

Internal viewer

Supported formats for the internal viewer: txt, ini, avi, tif, tiff, jpg, jpeg, dcx, eps, ico, pcx, pcc, png, ppm, pgm, pbm, tga, vst, afi, wmf, emf, bmp.

Supported formats for external HTML viewer: htm, gif, wmv, mpg,

External applications

You must install external applications if other file formats (file extensions) than the ones mentioned above are used (e.g. PDF files). The customer is responsible for the installation.

http links as document references

You can also pass http links to a browser for display without having to download a file beforehand. Use a path with the ”http” schema (paths are configured on the client via *System administration* > *System settings* > *Paths*).

These links are displayed using the internal HTML viewer provided by the *AIP*. The file extension does not affect the selection of the viewer.

Also the default browser configured in Windows can be used for the display. To do so, configure the following option in the “hytnrcfg.ini” file:

[Terminal->USR 0]
HTTPBrowser=standard

This setting is not recommended for an *AIP* with touch screen, because the operation of a browser can lead to problems.

5.13.3 Tools, Resources

The tab *Tools, Resources* displays the production resources and tools required for an operation.

Note: Documents are displayed separately in tab *Documents*.

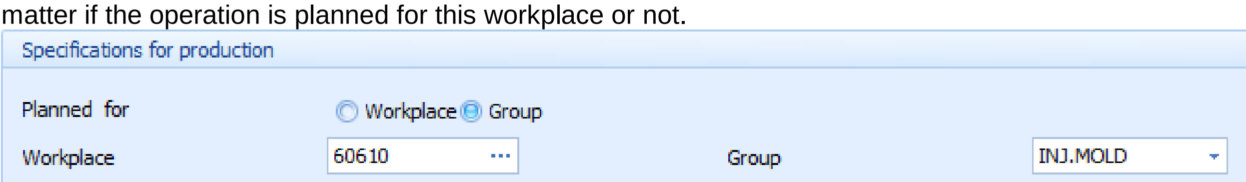
5.13.4 Components

The tab *Components* displays the material components required for the operation.

5.13.5 Progress

The tab *Progress* displays information on the status of the different operations of the order that includes also the currently the selected operation. The below-mentioned data is shown:

- Order
- Operation
- Operation designation
- Color of the status according to the control indicator of the status:
  S – (no color), V – gray, L – light green, U – yellow, E – green
- Status (text) of the operation
- Target quantity of the operation
- Quantity unit of the operation
- Yield that has been posted so far for the operation
- Quantity unit of the operation
- Workplace that is assigned to the operation according to the order management. It does not matter if the operation is planned for this workplace or not.



- Group the workplace is assigned to according to the workplace configuration.

5.13.6 Resource performance accounts

In tab *Resource performance accounts*, the following information is displayed for the resource performance accounts (RPA) 1 to 11 of the current operation:

- RPA abbrev.

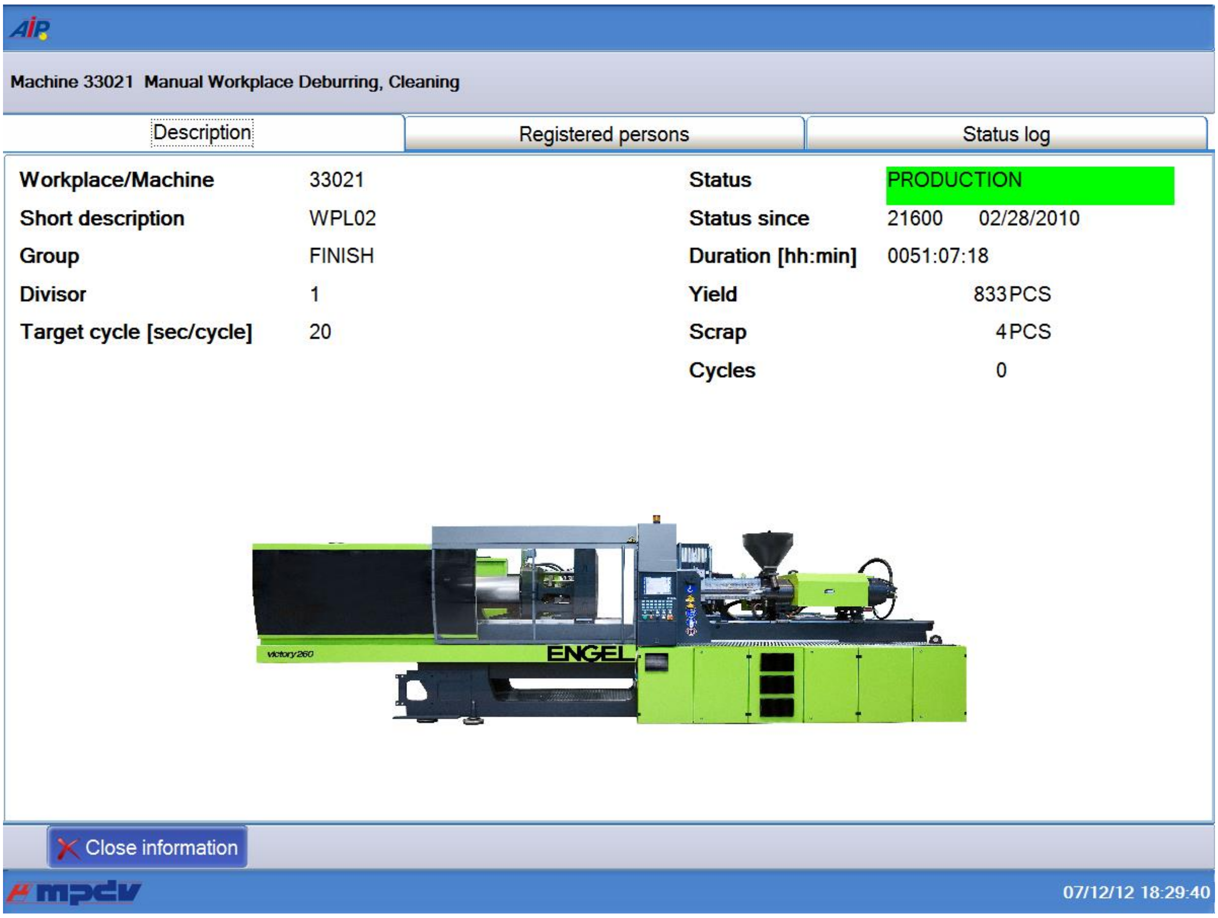
- RPA designation
- Posted duration in hours:minutes
- Total duration = total of all times of RPA 1...11

The durations are displayed in a graphic (to the right).

5.14 Machine information (machine info)

The machine information dialog provides the views and functions listed below.

5.14.1 Description



The tab *Description* shows information on the machine/workplace

Workplace/machine

Number of the workplace/machine according to configuration

Short name

Short name of the workplace/machine according to configuration

Group

Group the workplace/machine is assigned to according to configuration.

Partitioning

The current partitioning and the incoming cycles are used to calculate the number of parts per cycle posted for the workplace/machine.

Target cycle

Current target cycle used to monitor the workplace/machine.

Status

Current status

Status since

Point in time that specifies the beginning of the current status.

Duration

The duration specifies how long the current status has been available.

Yield

If the workplace has been configured as MDE workplace, this field shows the yield that has been posted in the current shift up to now.

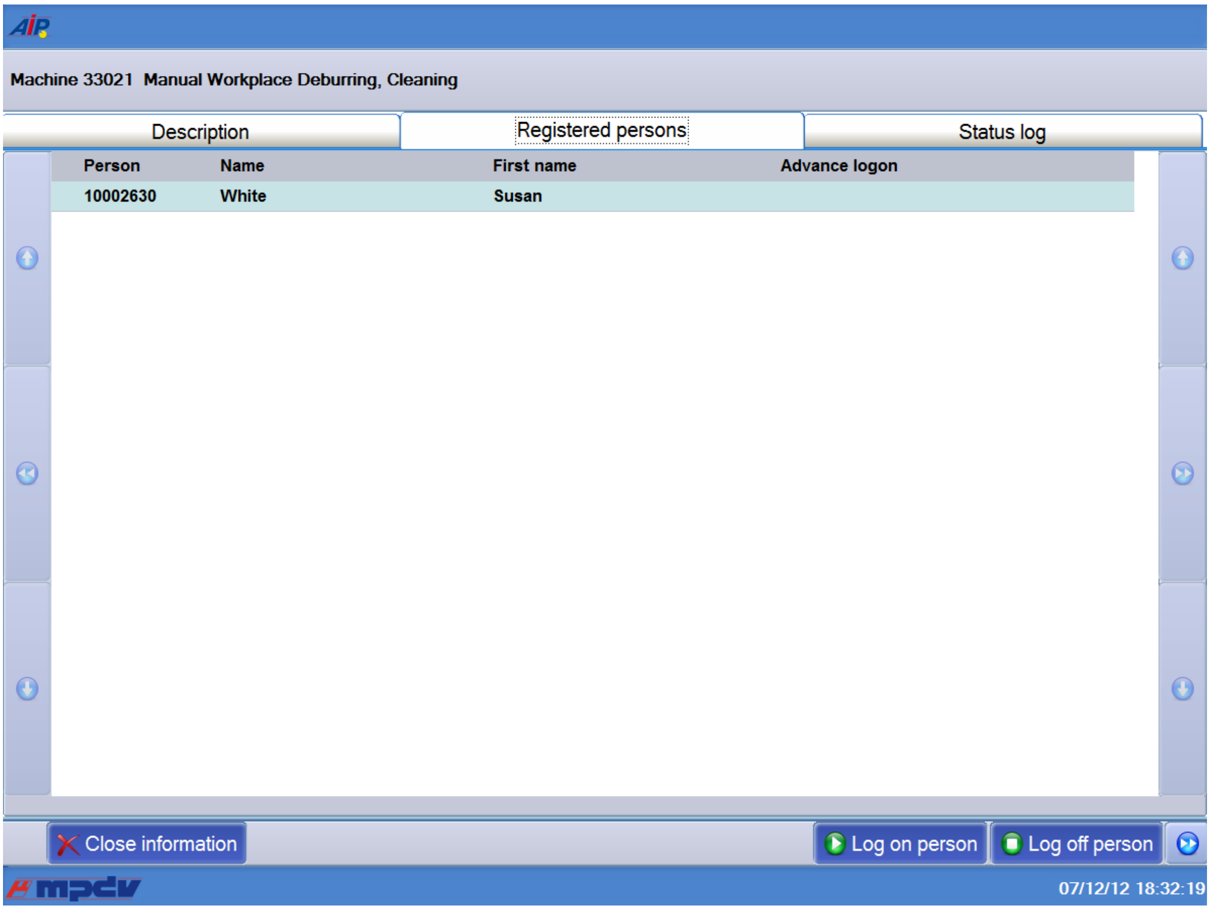
Scrap

If the workplace has been configured as MDE workplace, this field shows the scrap that has been posted in the current shift up to now.

Cycles

If the workplace has been configured as MDE workplace, this field shows the cycles that have been recorded in the current shift up to now.

5.14.2 Registered persons



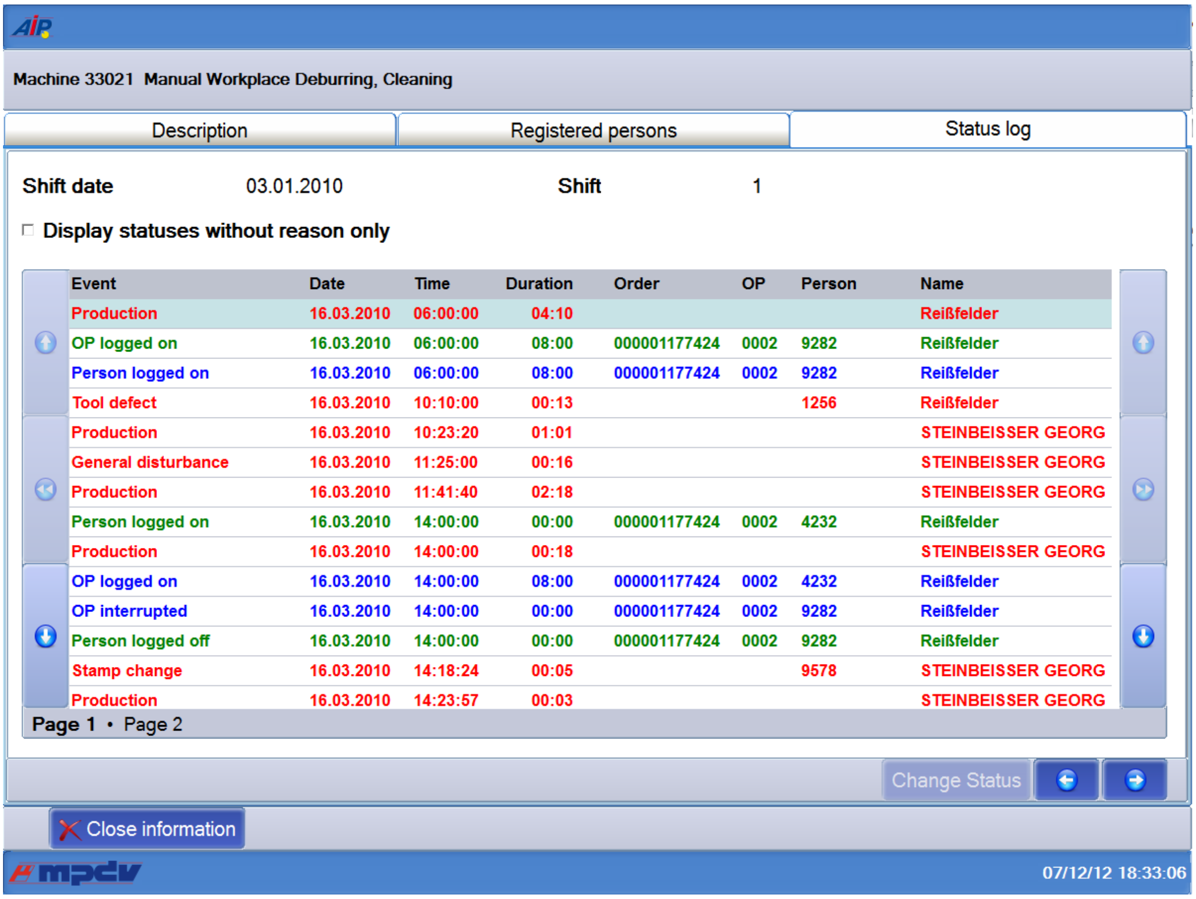
This overview shows the staff currently logged on to the workplace.

Use the following buttons to perform postings for staff:

- Log on person,
- Log off person or
- Log off everyone (on the next button page)

Click *Close information* to get back to the main view.

5.14.3 Status log



Event	Date	Time	Duration	Order	OP	Person	Name
Production	16.03.2010	06:00:00	04:10				Reißfelder
OP logged on	16.03.2010	06:00:00	08:00	000001177424	0002	9282	Reißfelder
Person logged on	16.03.2010	06:00:00	08:00	000001177424	0002	9282	Reißfelder
Tool defect	16.03.2010	10:10:00	00:13			1256	Reißfelder
Production	16.03.2010	10:23:20	01:01				STEINBEISSER GEORG
General disturbance	16.03.2010	11:25:00	00:16				STEINBEISSER GEORG
Production	16.03.2010	11:41:40	02:18				STEINBEISSER GEORG
Person logged on	16.03.2010	14:00:00	00:00	000001177424	0002	4232	Reißfelder
Production	16.03.2010	14:00:00	00:18				STEINBEISSER GEORG
OP logged on	16.03.2010	14:00:00	08:00	000001177424	0002	4232	Reißfelder
OP interrupted	16.03.2010	14:00:00	00:00	000001177424	0002	9282	Reißfelder
Person logged off	16.03.2010	14:00:00	00:00	000001177424	0002	9282	Reißfelder
Stamp change	16.03.2010	14:18:24	00:05			9578	STEINBEISSER GEORG
Production	16.03.2010	14:23:57	00:03				STEINBEISSER GEORG

At first, the table shows the following events of the current shift:

- All machine/workplace statuses
- Production lock (set manually, removed manually)
- Postings relating to orders (OP logged on, OP interrupted, OP logged off)
- Postings relating to staff (person logged on, person logged off)

Use the buttons  and  to scroll back and forth shift by shift. You can configure for the specific terminals how many shifts you can scroll back and forth. Use the following option in the hytnrcfg.ini (3 shifts by default):

[MDE->Options 0]
MSTAT.SKNRRANGE=3

In the above example, the screen shows shift 1 if you click in shift 3. Scrolling back has the same behavior.

Subsequent assignment of reasons

If the machine monitoring function is active, the status switches to the “not assigned” status in case of an absence of signals. When the malfunction has been removed, the machine immediately switches into the production status. The operator cannot enter a reason for this status.

In the status log, you can now list these statuses without reason and subsequently assign a reason to the statuses.

Statuses without reason are statuses that the system has automatically set from "not assigned" to "production" in the course of an active machine monitoring. The production indicator “general disturbance” is assigned to this status and the relevant time is posted for this status.

If an operator manually sets a status, this status usually has a reason assigned, also if the operator uses the production indicator “general disturbance”.

If you enable the checkbox *Display statuses without reason only*, you can filter the list and directly show the statuses without reason only (i.e. automatically assigned reasons). These statuses can then be changed. Note: When you have changed the selection, you must reload the list using the green arrow button!

The *Change status* button gets active when you select a status without reason. You cannot change statuses set manually by the user. The button is grayed out (disabled).

If you click the *Change status* button, you can subsequently assign a reason to this status.

Note

Restriction: Only the MDE data (log records) is changed when you subsequently assign a reason. Postings of other objects (e.g. BDE log records) are not changed.

You cannot subsequently assign reasons to "Short-term disturbances/malfunctions".

5.15 Merged operations

General information

You use these functions to create or to end "**merged operations**" (MOP) on the AIP terminal. A merged operation includes a group of single operations. The system records the time for the merged OP and proportionally distributes the time to the single OPs. Each person can manage a maximum of one merged operation.

You require a license to use the functions for merged operations on the AIP.

To merge operations, you must click the button *Log MOP on* in the workplace screen. Merged operations are logged off and interrupted like single operations using the *Interrupt operation* or *Log operation off* buttons.

With merged operations created on the terminal, you can only perform logon/logoff/interrupt postings on the terminal. The same also applies to the single operations included in a merged operation. Postings on the MOC are not possible.

Also note that there are some restrictions for merged operations that are described in the documentation MBL_CollectiveOperationProcessing.pdf. Not all postings are possible that are possible with normal operations.

5.15.1 Log merged operation on

You can combine up to 20 operations to form one merged operation. On the *AIP*, only one operation (the new merged operation) is then displayed in the list of running operations..

Posting process

Select a workplace, before you log on a merged operation. When you then call the dialog, the workplace field is already populated.

Calling the function *Log MOP on*

Click the *Log MOP on* button in the workplace dialog. The user is navigated through the dialog.

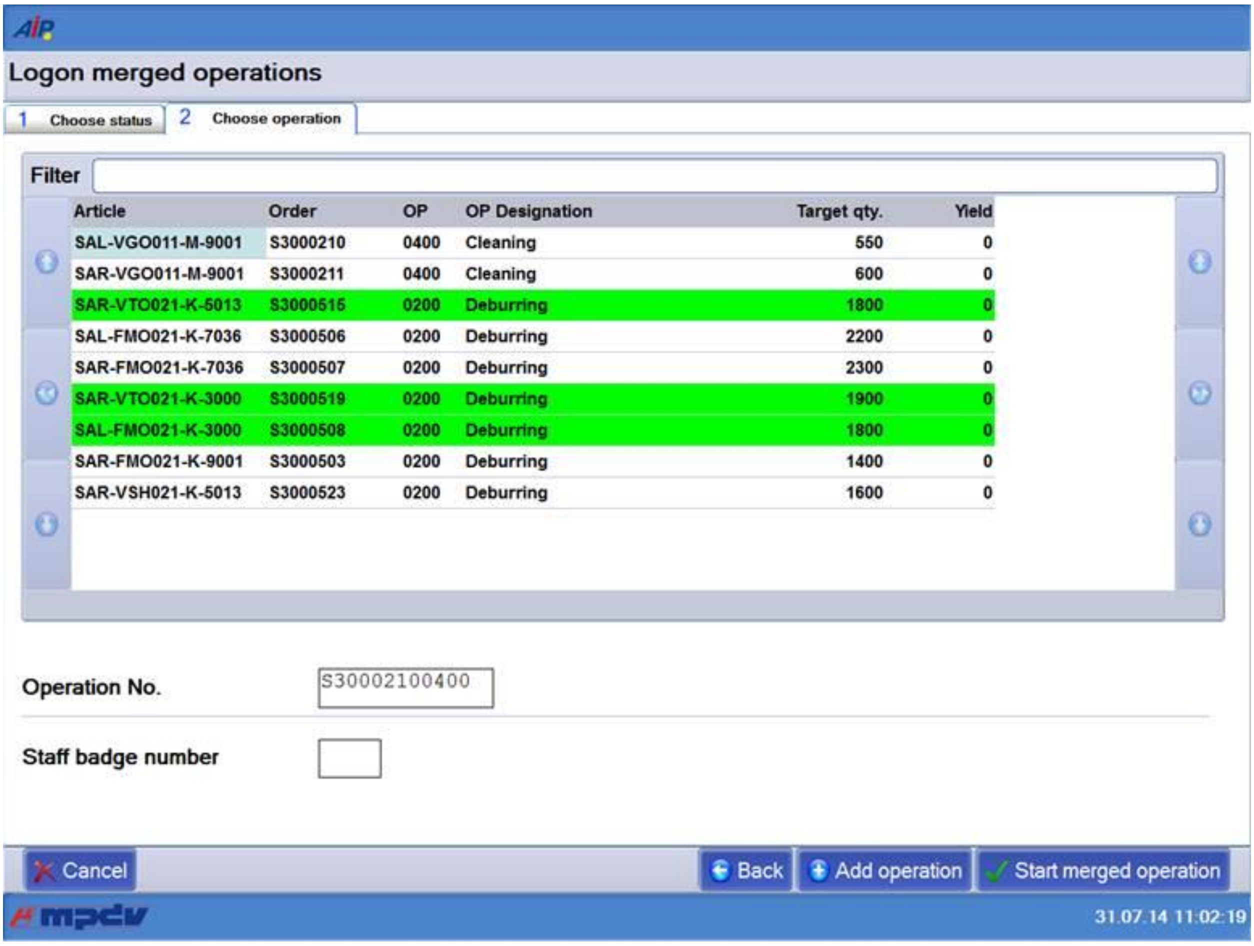
Choose status

Enter the status number. Optionally, you can also enter the estimated duration and a comment.

Choose operation

Select all single operations from the list that you want to integrate in the merged operation. Manually select the operations in the list as usual.

When you have selected the single operations, click the button *Add operation*. All selected operations are displayed in green in the list:



Staff badge number

Enter the staff badge number here. With merged operations, the person is always logged on with the operation, i.e. a separate staff logon is not possible.

Confirmation via *Start merged operation*

The specified operations are immediately logged on. If the terminal is ONLINE, the system makes a validation check for all entries.

When the merged operation is successfully logged on, an entry in the order overview is generated. This entry includes the characters “**SAM-**“ for merged operation and the badge number (e.g. **0160**). The merged operation of the example would therefore be: **SAM-0160**.

5.15.2 Interrupt/log merged operation off

You interrupt or log off a merged operation like single operations. Select an operation named **SAM-<badge number>** from the list of running operations.

Posting process

Select the workplace and the merged operation that you want to log off.

Calling the function

Click the button *Interrupt operation* or *Log operation off*.

Log merged operation off

Enter the badge number of the person who performs the logoff.

Choose status

Enter the status number. Optionally, you can also enter the estimated duration and a comment.

Choose operation

If the entry of quantities is configured for merged operations in the terminal configuration, a list of the different operations is displayed. To upload part quantities, you can select the relevant single operation, enter the required quantity and click the button *Partial confirmation*. The system does not immediately upload the quantities entered here. The quantities are only uploaded when you confirm the dialog by clicking the button *Interrupt/Log off merged operation*. If the dialog is canceled (ESC key or *Cancel* button), the dialog is closed and no data is uploaded. The partial confirmation is discarded.

Confirmation of the dialog *Log off/Interrupt merged operation*

The merged operation is logged off or interrupted when you confirm the dialog.

If the order is properly logged off, the merged operation is unmerged. After interruption, the single operations are available again in the sequencing list.

The quantities are posted according to the configuration and using the part quantities uploaded.

Further notes

- If you do not enter a quantity with the function *Interrupt merged operation*, then no quantities are posted. With the function *Log merged operation off*, the target quantity is booked as yield quantity.
- If you interrupt a merged operation, the system interrupts all single operations assigned to the merged operation. As a result, the sequencing list does not include the merged operation after interruption, but the single operations. If you want to log on a merged operation again, you must again select or assign the single operations.

6 Specific Features of Machine Data Collection

The functions described in the paragraphs that follow are only active at workplaces that fulfill the following requirements:

- The machine/workplace has been configured as "single workplace"
- The machine/workplace is assigned to a terminal, which is configured with operation mode "MDE".

If the AIP2 is in use, machine data collection and its functions are completely performed via PCC. The following functions are visualized by the AIP2, but triggered and executed by the PCC.

These functions are not available for group workplaces.

To ensure proper processing and posting, terminals with "MDE" operation mode must not be switched off during times without shift.

6.1 Shift automatic

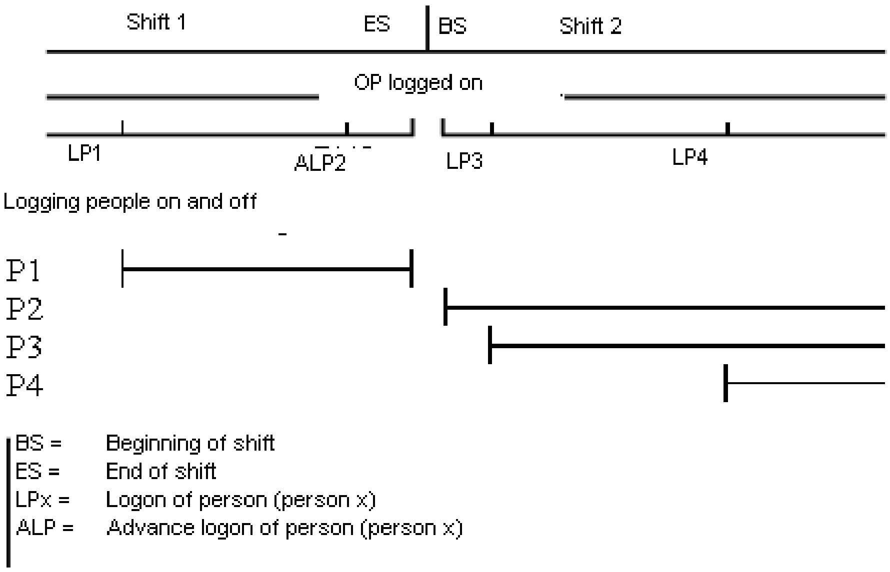
A shift model has to be assigned to each workplace/machine. Due to the information given by this shift calendar, the PCC is able to identify automatically the beginning and end of shifts for its assigned machines.

The shift automatic option activates functions facilitating data collection and operability:

- the OP that is logged on is automatically interrupted at the end of the shift.
- if configured accordingly, this OP is logged on again when the next shift starts.
- staff can log on in advance to a terminal within a specified period of time before the beginning of the shift. When the shift starts the next time, the terminal logs them on to the OP. This period of time is defined in the *Terminal configuration* (option: *Waiting period for advance logon of staff*).

You must perform changes to the shift model at an early stage because the PCC reads the shift data only at cyclic intervals from the HYDRA server (by default every 50 minutes).

The following diagram shows a time flow of logons and logoffs during a shift change.



Description of the process

1) An OP was logged on during shift 1 and its production continues in the following shift 2.

2) Person 1 logs on to the workplace.

3) Person 2 arrives shortly before the shift ends and logs on to the workplace. Since the logon takes place within the 30 minutes of advance logon time, the terminal recognizes an advance logon.

4) The registered OP is automatically interrupted when shifts end and staff logged on is logged off.

5) The OP interrupted beforehand, is logged on again when the shift starts. Staff logged on in advance is still logged on.

6.2 Machine monitoring in general

If you connect a machine to a shop floor terminal, the automatic data collection is possible.

- Collection of production times and downtimes
- Direct assignment of malfunction reasons by collecting operating signals from machine control systems
- Collection of quantities in different quantity accounts (yield, scrap, rework, open quantity)

If you want to use the machine monitoring function via a machine signal, you must first define relevant machine statuses. In addition to the production status, all status reasons that are relevant for the HYDRA collection are created for a machine. The user may configure them individually. By default, HYDRA only displays and posts one status at the time of consideration. Only this way, you can ensure that production times and downtimes are considered in the correct chronological order.

The monitoring type is configured in the Workplace/machine configuration (tab *MDE configuration* > *Monitoring type*). The following values can be set:

Cyclic monitoring	monitoring cycle times
Monitoring via operating signal	Monitoring operating signals
No monitoring	Here, it is possible to define a new machine status at all times. Also the “Production” status must be assigned manually on the terminal.

6.3 Monitoring of cycle time

6.3.1.1 Terms and definitions

Target cycle

The target cycle is a planning value specifying the pulsing of a machine to produce an article. Its value may depend on different factors, e.g. on the tool or material to be used.

The target cycle does not depend on the number of produced parts. In HYDRA it is recorded and processed as duration per 1000 machine cycles.

For the cycle time monitoring, the value of the target cycle is used as default setting for the production of the operation.

Cycle extension

The cycle extension is a percentage to extend the target cycle. It can be entered in a range from 0 to 5000. It is used to balance possible fluctuations during machine monitoring. The cycle extension is offset against the target cycle. A value less than 100 is a shortened cycle; a value greater than 100 is an extended cycle.

The *Cycle extension* is configured in the *Workplace/machine configuration* (tab *MDE configuration* > Cycle extension).

Minimum cycle time

The minimum cycle time is a kind of “minimum value” used for monitoring cycles, if the product of target cycle (set manually on the terminal or implicitly with OP logon) multiplied by cycle extension is less than this minimum cycle time. Thus, the minimum cycle time is configured at the machine and independent of the article to be produced or tool to be used.

The *Minimum cycle time* is configured as a value in seconds in the *Workplace/machine configuration* (tab *MDE configuration* > *Monitoring* > Min. cycle/disturbance time).

Minimum cycle time: Special features when using DS-100

If you use DS-100, the minimum cycle time is adapted to the technical communication possibilities. Also the number of connected DS-100 devices to a terminal is considered.

The system (PCC) identifies the number of connected DS-100 devices and calculates the so-called *DS-100-Minimum cycle time*:

*DS-100-Minimum cycle time [sec] = <number of connected DS-100> * DS-100-factor*

If the *DS-100-Minimum cycle time* is longer than the configured minimum cycle time, the *DS-100-Minimum cycle time* is then used to calculate the default cycle time.

The *DS-100-factor* is by default: 1.0 [sec]. If necessary, it can be adapted in the mdeb2.ini, e.g.:
[INIT]
DS-100_MACHINE_COUNTCYCLEFACTOR = 0.8

Example

Five DS-100 are connected to one terminal.
The minimum cycle time stored in a workplace is 2 seconds.

DS-100-minimum cycle time = 5 x 1.0 second = 5 seconds

As the DS-100-Minimum cycle time is longer than the minimum cycle time, a minimum cycle time of 5 seconds is then used in processing. This time is now used to define the default cycle time.

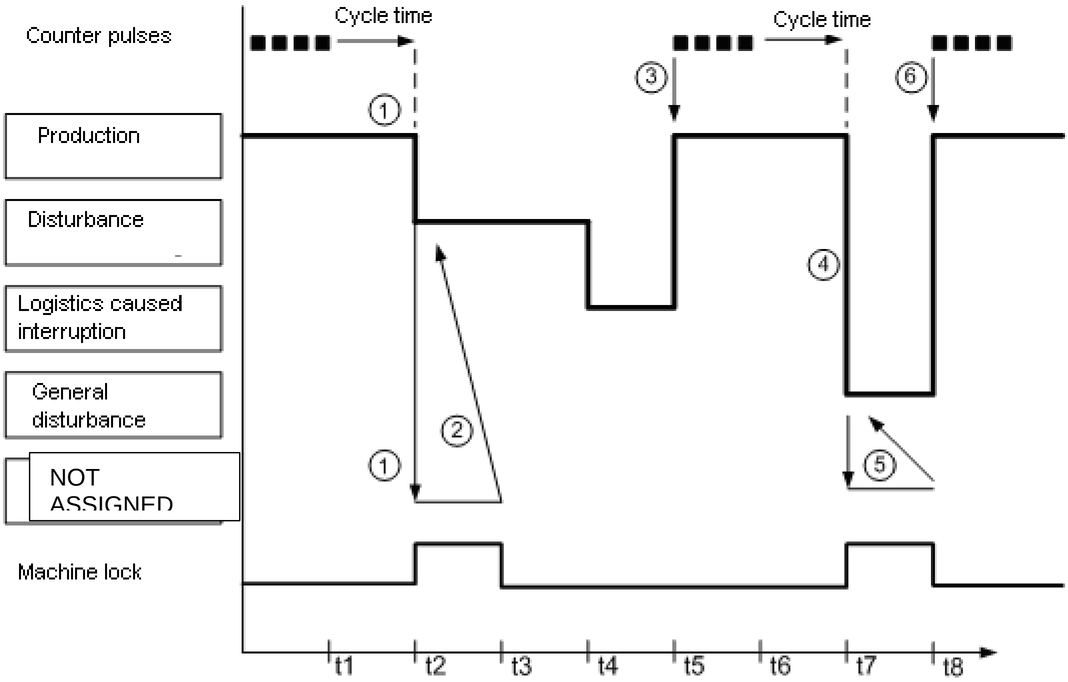
6.3.1.2 Definition of the default cycle time for the machine monitoring

The longest target cycle is always determined from all OPs logged on to a machine for the collection of cycle times: max($SZY_{OP1..n}$)

Afterwards it is checked whether this target cycle multiplied by the cycle extension is greater or equal to the minimum cycle time. If this is the case, this target cycle is multiplied by the cycle extension and used as monitoring cycle. If, however, the minimum cycle time is greater, this minimum cycle time (without cycle extension) is used as monitoring cycle.

If both, the minimum cycle time and the target cycle stored for the operation, are 0, the default cycle time is set to 60000 seconds [per 1000 machine clocks].

6.3.1.3 Processing



If the machine transfers counter pulses to the PCC via an interface, the PCC automatically switches to the status "Production".

In the event counter pulses fail to appear, the PCC waits for the default cycle time z and sees if the machine still sends a counter pulse. If this is not the case, the PCC interprets this as a malfunction and changes the status of the machine into 30000 “not assigned" [1]. This status refers to a malfunction that has not been confirmed.

If the user now confirms this malfunction by assigning a specific status [2], the length of time since counter pulses failed to appear is posted back to this status.

If the machine restarts to send counter pulses to the PCC via an interface [3] [6], the PCC automatically switches to the status "Production".

If the operator did not assign a reason to this malfunction and if the PCC switches back to production because of signals [6], this length of time without malfunction reason is posted back to the status "General disturbance" [5].

If the machine constantly switches between the "production" and the "not assigned/general disturbance" status, you must check the machine connection or the combination of target cycle and cycle extension. If necessary, you must adapt the minimum cycle time.

You can only enter or change a status, if the active status is not "production".

If the operation with the longest target cycle is interrupted or finished, the operation with the next longest target cycle is used for collecting cycles. In this case, the logic described beforehand applies again. Once the last operation is logged off, the last used target cycle is kept as target cycle.

6.4 Monitoring of operating signals

6.4.1.1 Terms and definitions

Minimum malfunction time

If you select the monitoring type *Operating signal*, the minimum disturbance time is specified in this field.

The minimum disturbance time is a time in seconds. It defines the time that a malfunction/disturbance must continue before the machine changes from the status "Production" to the status "Not assigned".

If operating signals are monitored, the status is directly changed. You can use the following explicit option in the MDEB2.ini to disable this behavior (deactivation of direct status change). Result: the status is only changed when the **minimum disturbance time** has expired:

MDEB2.INI

```
[INIT]
;Activating the direct status change (globally or for a specific machine)
SetMStatusDirect=1
SetMStatusDirect@<machine number>=1

;Deactivating the direct status change (globally or for a specific machine)
SetMStatusDirect=0
SetMStatusDirect@<machine number>=0
```

6.4.1.2 Processing

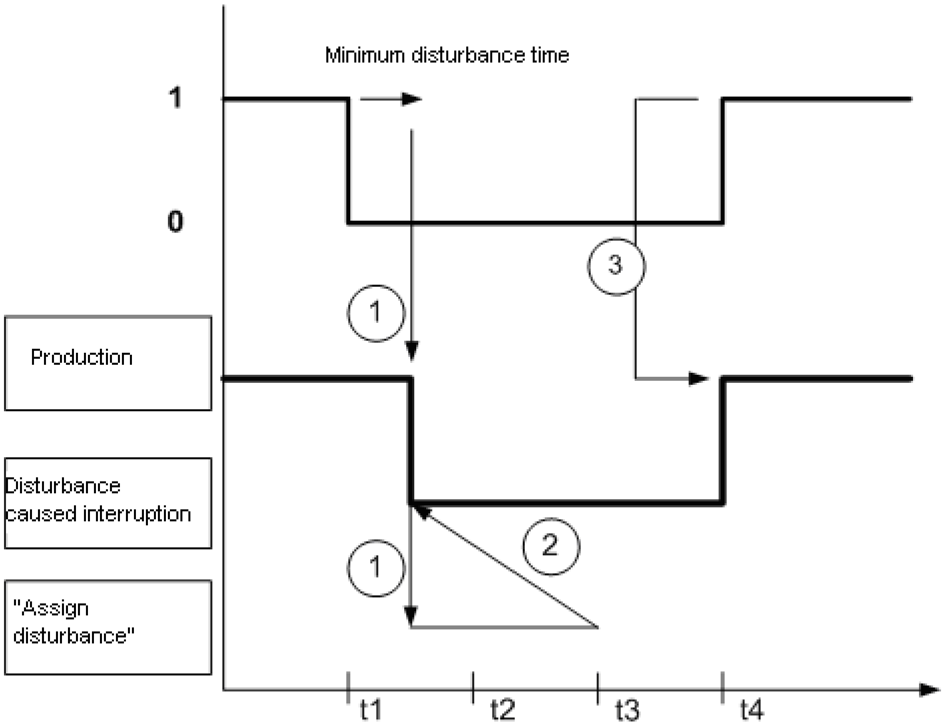
If the machine sends an operating signal via the interface, the PCC automatically switches to the status "Production". The input for the operating signal must be defined in the status assignment (menu: Master data → Workplaces/Machines → Status assignment, tab Status change, option Automatically via digital input).

Processing with direct status change

If this signal is no longer available, the PCC interprets this as a malfunction and changes the status of the machine to 30000 “not assigned" [1]. This status refers to a malfunction that has not been confirmed.

Processing with deactivated direct status change

ctwin_Zeichnung_16.06.2005_II.VSD



If this signal is no longer available, the PCC waits for the minimum disturbance time d. If the signal is still not available, the PCC interprets this as a disturbance and changes the status of the machine to 30000 “not assigned" [1]. This status refers to a malfunction that has not been confirmed.

If the operator now confirms the malfunction by assigning a specific status [2], the time since operating signals have not been recorded ([1]) is posted back to this status.

If the machine restarts to send signals to the configured input [3], the PCC automatically switches to the status "production".

If the operator did not assign a reason to this malfunction and if the terminal switches back to production because of signals, this length of time without malfunction reason is posted back to the status "General disturbance".

If the machine constantly switches between the "production" and the "not assigned/general disturbance" status, you must check the machine connection. If necessary, you must adapt the minimum disturbance time.

You can only enter or change a status, if the active status is not "production".

6.5 Lock production status (production lock)

During machine monitoring, the function *Lock production status* prevents the status "Production" from being automatically set because of an incoming production signal (clock pulse, operating signal). The production lock can also be used to determine whether and how quantities (counter readings) are posted during this time.

Consequently, the status set here overrides the production signal until the production lock is manually deactivated on the terminal. If the machine sends new signals, the PCC does not automatically switch to the status "Production". Thus, switching to “production” is disabled (“production lock”).

If a workplace/machine is unable to switch to the “production” status, the terminal visualizes this by displaying an exclamation mark ("!") in the "machine information" dialog.

In this case, the current status remains – despite the machine pulses received.

All items produced during the locked status are either posted as yield, scrap or not at all. That depends on the configuration of the workplace.

The production lock may be enabled or disabled explicitly by clicking the button "Lock production status".

If the workplace/machine status is configured accordingly (menu: Master data > Workplaces/machines > Status assignment), the production lock may also be set automatically, i.e. along with setting a status.

Authorization check when setting the production lock manually

You can configure that only if a relevant authorization is available, the person is allowed to manually (explicitly) set or remove the production lock.

For this purpose, the dynamic dialog M_PSPERRE must be activated (customizing). If this dialog is activated, you must enter the staff badge number.

If the dialog is active, it opens when the operator clicks the button “lock production status”. Once the badge number has been entered, it is checked whether this person is authorized to activate/remove the production lock. The system checks, if the option *Change of production lock* in the HR master data (tab *Shop floor data*) is enabled.

If the terminal is OFFLINE, the terminal configuration specifies the behavior (option *Checking required*).

Logging of the production lock

The manual activation/deactivation of the production lock is recorded as an event in the server and may be evaluated in the machine history.

.

But if the production lock is enabled or disabled implicitly by a status change, no specific event will be recorded and, as a result cannot be evaluated in the client.

In case a machine is locked for production and the *AIP2* terminal together with the PCC are restarted, the production lock is automatically disabled after the restart. The changed production lock is not logged.

6.6 Machine lock

You can define for each status whether it should trigger a “machine lock”. This can be configured (*Status assignment* > tab: *Control* > *Set machine lock output*).

If you set a machine lock, an output is being set, which may trigger a relay. In this case, the logical output is defined in the *Workplace/machine configuration* (tab: MDE configuration > inputs/outputs > machine lock).

Notes on using the machine lock on DS-100 terminals

The value “1” has to be entered in the field “Machine lock” of the *Workplace configuration* (tab: MDE configuration > Inputs/outputs), provided that the relay is to be set for the DS-100 terminal.

The relay is set for all statuses assigned to the "machine lock" as well as for the "not assigned" status.

6.7 Output "Target quantity reached"

An output may be set to trigger a lamp via a relay (to be provided by the customer), when the target quantity of the currently registered operation has been reached.

The logical output is defined and configured in the menu: *Master data* > *Workplaces/machines* > *tab: MDE configuration* > *Inputs/outputs* > *Target quantity reached.*

The target quantities of a machine are checked:

- after restart
- after changing the target quantity using the corresponding posting dialog on the AIP2 terminal
- once an operation has been logged on, off or interrupted on the AIP2 terminal
- after posting manual quantities (partial confirmation) on the AIP2 terminal
- after local quantity events of the MDE product group (automatic quantities)

If several operations are logged on to a workplace/machine at the same time, the logical output is set as soon as the target quantity has been reached for one of the orders. The signal will be reset if this OP is interrupted.

6.8 Scrap reasons depending on status & production lock

Two scrap reasons can be defined for a status. One scrap reason is applied, if the production lock is activated (see section *Lock production status (production lock)*), the other scrap reason is applied, if the production lock is inactive. If a scrap reason is defined for the status currently available at the machine, the counted scrap will be posted with this specified scrap reason.

If a scrap reason is configured at the counting input, this one takes priority. Consequently, if a specific scrap reason is assigned to a counter, this scrap reason takes priority even if another reason is configured for the currently available status.

6.9 Setting of outputs depending on status and posting scenarios

By way of an advanced configuration, digital outputs of the machine may be set if a specific machine status and a defined posting scenario coincide. Further details on this configuration can be found in the document entitled *MDE_digital_output_depending_on_scenario.pdf*.

Please note the following for the local administration of the machine status in the terminal:

If status changes are posted using SMA, PDM or on another terminal, the machine status is only transferred when reloading the machine list, if this has been configured explicitly via the "FollowExternStatus" option:

HYTNRCFG.INI
[Tnr Konfiguration 0] FollowExternStatus=on

When the status is changed by reloading the list, the terminal checks the configured logical outputs and sets or resets them, if required.

6.10 Manually set status vs. automatically set status

A manually and an automatically set status result in different system behaviors.

A status that is set manually takes priority to a certain degree and cannot be overwritten by another status if an input signal is available, for example. The only exception is status *Production*: This status also overwrites the manually set status. A status that is set automatically can be overwritten by an input signal (irrespective of the status).

The information whether a status is set manually or automatically is saved and remains available.

Special feature with status 999: If a status 999 was set manually before shift change, then the system changes this status to an automatically set status 999 with shift change if this is activated for the day type (weekend automatic). The behavior after shift change is then the standard behavior that applies with an automatically set status.

7 Barcode Input

A barcode is machine-readable information, which may also directly be generated within the shop floor system. The principal reading devices used are: barcode scanners, swipe card readers or scanners. By default, the terminal supports the barcodes "39", "128" as well as ”interleaved 2 of 5 "

Only barcode readers connected to the serial interface (COM interface) can identify barcodes and assign them automatically to the respective input fields at the terminal. This is impossible for barcode readers “connected” via keyboard.

Barcode structure for operations

AAAAAAAAAAAAGGGGG

Place	Name	Min. length	Max. length *	Example
*	Asterisk	1	1	*
A	Order number	1	12	8
F	Sequence number	0	2	0
G	Operation number	1	4	3
S	Split number	0	2	0
*	Asterisk	1	1	*
Length of BDE order barcode without asterisk (example)				11

* Depends on the configuration of field lengths in HYDRA's basic parameter settings

Barcode structure for machines/workplaces

MMMMMMMM

Place	Name	Min. length	Max. length	Example
*	Asterisk	1	1	*
M	Workplace/ machine number	8	8	8

Place	Name	Min. length	Max. length	Example
*	Asterisk	1	1	*
Length of machine/workplace barcodes without asterisks				8

To identify a barcode as a machine/workplace number, the barcode must be 8 characters long. If the machine/workplace number is set to “numerical” in the basic parameter settings, it has to be filled up with leading zeroes to reach 8 characters. In case of alphanumeric machine/workplace numbers, they have to be filled up with underscore characters ("_") to the right.

Barcode structure for machine statuses

NNNNN0

Place	Name	Length
*	Asterisk	1
N	Machine status, with leading zeroes	5
0	Always "0"	1
*	Asterisk	1
Length of status barcodes without asterisks		6

8 AIP configuration of barcodes

8.1 Configuration in ctaip.ini

Readers are configured in section [comports] of the file ctaip.ini:

Section [comports]	
COM1=0 COM2=0 COM3=0 COM4=0 COM5=0 COM6=0	Possible initializations of comports COMx=0 => is not used (by default) COMx=BAR COMx=PSTD PLEASE NOTE! Only starting from reader firmware 69355E COMx=LEGIC COMx=PLG COMx=RFLESER COMx=MSS COMx=SLGB Entry for readers of the type "Schlagbaum" COMx=UKEY U-Key readers Byte 12 and 13 set up the badge/ID card number Byte 14 and 15 set up the company number e.g.: 01010D01 | 0E020000 | 04000003 | E7175600 03E7 ➔ badge number ➔ 999 1756 ➔ company number ➔ 5974 COMx=KABALEG Kaba Benzing Legic **With Bedanet 9580 always COM4** 7 bytes are transferred as of byte 15 XXXXFFFFKKKKKK F:= company number K=badge number COMx=MBB-S6 MBB-S6 reader **Attention!** Please note! This reader type requires the RS-485 converter to be modified (ECHO=OFF RTS=High) Comports masking has to be set for CLEA + DEUT MBB-S6-MASK=XXXXXXXXXXKKKKKK COMx=DRV_UCR New LEGIC Advant **PZE/MF** reader. required for new PZE readers. The names of these readers include "LGA". **Please note:** If badges/ID cards are used not complying with MPDV's standard ID, the following parameter must be used for masking in section [Comports-Mask]: DRV_UCR-MASK=.... COMx=PLG-CRYPT New LEGIC Advant **ZKS** reader required for new ZKS readers. The names of these readers include "LGA". It might be necessary to customize the file plg_crypt.ini for specific customers Set this parameter if Kabalegic and badges/ID cards with MPDV ID are used

	KABALEGM=ON By default, the Kaba Benzing badge ID is used. In this case the parameter is to be commented out As of ctaip #V 2.0.2.27 Read out Kabalegic using configurable search string (e.g. 4F) KABALEG-SEARCHSTRING=XX Correct configuration takes priority (2 places, values 0..9,A..F)

Section [Comports-Mask]	Example
LESERTYP+'-MASK' =Masking	applicable abbreviations for masking 'T' = Telegram number 'L' = Reader number 'F' = Company number 'K' = Badge/ID card number 'E' = Replacement number 'P' = Check digit (not implemented) otherwise e.g. 'X' = Placeholder (character to be ignored)
Examples: SLGB-MASK=TTTTLLFFFFFFKKKK BAR-MASK= MBB-S6-MASK=XXXXXXXXXXKKKKKK	Masking for readers of the type "Schlagbaum": TTTTLLFFFFFKKKK = Data string of the reader The badge number may be recorded at every position of the data string Please note!! Impossible for PLG // Status 15 October 2003 DB

The below section has been designed for the configuration of (PLG) Polling Legic Readers or (PSTD) Polling Swipe Card Readers. These readers are mostly used for PZE/ZKS terminals (CT-380) or ZKS terminals (CT-385).

Section [init]	

PlgTimeOut=2 FreischaltungZyklus=3 MaxComError=15 UngueltigZyklus=300 InitComError=50 MSSImmediateWrite=false Leser-4-Hupe-Aktiv=true LnrErrorPCnt=100 PinCodeLen=4 ComResetTimer=5	→ TimeOut when starting polling → Cycle for activating/releasing access → Maximum number of communication errors until access is in "malfunction" → Cycle for sending the status "invalid“ (invalid badge) in order to re-initialize the reader → Number of communication errors in order to re-initialize Comport → ZKS immediately activates channels for MSS connection (true = not efficient) → Enables an acoustic signal (buzzer/horn) for the status "valid" (valid badge) → Polling is only performed with every n-th attempt if the reader is in the status "malfunction=e". (priority polling) → Length of PIN code input → Minimum cycle for comport re-initialization
ReInitComport=true	Activation of automatic comport re-initialization → by default = false (= without automatic re-initialization)

The below section has been designed to configure MBB readers triggered via a thread in a DLL. These readers are only used for ZKS terminals (CT-385).

Section [MBB-S6-configuration]	
MinMSecDauerOeffner=200	→ MinMSecDauerOeffner = Minimum duration "door opener" in MSec -> by default [200]
;****************************	;*** MBB-S6 - system settings (do not change) *******
ProcessMessages=true	→ ProcessMessages = Processing of Windows messages while waiting for the application result -> by default [true]
ThreadSleep=20	→ ThreadSleep = Sleep in MSec of MBB-DLL while waiting for application result -> by default [20] range of values [10 .. X] -> changes might lead to increased CPU loads or bad response times

8.2 Configurations in hytnrcfg.ini

Entry	Comment
Section [Terminal->USR 0]	
SuppressBarcodeError=On	Suppresses messages such as 'Barcode ... is wrong', etc. Required if the barcode is processed in the script and normal identification processes cannot identify the barcode
OnBarcode=P_AN	If the AIP basic screen is opened and a barcode is read in, the dialog P_AN opens instead of the dialog M_MST. The scanned badge/ID card number is transferred.
BarcodePrefixChar=$	Configuration of an alternative separator for barcode prefixes. Can be used if a dot not used as prefix identifier is included at the third place of a real barcode.
ON-KNR-CODE=<Dialog> (By default "M_INFO.PERS“)	As of V# 2.0.2.57 Configures the dialog that opens from the main dialog when scanning the staff badge/ID card number using readers such as Legic, Kabaleg, UKey, .

8.2.1 Notes/configurations for concurrent lengths

Entry	Comment
Section [terminal configuration 0] and/or [terminal configuration 2XXX];	**(general configuration) (2XXX terminal-specific configuration)**
ANR-COMPLETE-BARCODE-ONLY=TRUE	Option to avoid processing of incomplete order barcodes. (e.g. AUNR,AUNR+AFOLG,…) -> The option is set to <FALSE> by default

RIVAL-BARCODE-LEN-INFO-MSG=FALSE	Option disabling messages indicating that barcodes cannot be processed due to concurrent lengths. -> The option is set to <TRUE> by default

Please note for configuration: the section may only be inserted provided it does not yet exist.

Sample configuration (for all terminals)

```
[Tnr configuration 0]
ANR-COMPLETE-BARCODE-ONLY=TRUE
BARCODE-LEN-INFO-MSG=TRUE
```

The above configuration prevents incomplete order barcodes from being processed at all terminals (configuration for one terminal with [TNR configuration 2xxx] xxx=terminal number including leading zeroes "0"). If this option is set, only complete order barcodes will be processed. Consequently, valid lengths for order barcodes result from added partial order lengths *(Please note: //- ANR (order number) parts with (*) may have the length '0')*

➔ AUNR + AFOLG *(*)* + AGNR + UAGNR *(*)*
➔ AUNR + AFOLG *(*)* + AGNR + UAGNR *(*)* + SPLNR *(*)*

The following message is shown if a barcode cannot be processed due to concurrent lengths:

Note [concurrent length]

A barcode (<VAL>)<n>whose field assignment (<IDS>)<n>cannot clearly be defined, has been scanned.<n><n>Processing is only possible<n>if one of the fields indicated is focused.

9 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
--------	--------------------	---- General ---
00.	00.ABC123	Data not defined, ➔ 00. will be deleted and "ABC123" will be passed on to standard processing
01.	01.OK 01.ESC	Action barcode ➔ Dialog cancelled or ended with OK button or Esc button.
--------	--------------------	---- HYDRA-ADE + HYDRA-LLE + HYDRA-MDE ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Sub-order number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16,200	Machine → Acronym <MNR> ➔ Passed on to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> ➔ Passed on to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> ➔ Passed on to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > ➔ Passed on to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> ➔ Passed on to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
--------	--------------------	----HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL ---
40.	40,100 40.MONTAGE	Destination → Acronym <ZLO> ➔ Passed on to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> ➔ Passed on to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→➔
43.		Throughput batch number → Acronym <DLL>→➔
44.		Alternative batch number → Acronym <CNR:ALT1>→➔
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
--------	--------------------	---- Mainly for the HYDRA-PZE module ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> ➔ Passed on to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> ➔ Passed on to dialog with FGR=1 or FGR=1001
--------	**--------------------**	**---- Customer-specific barcodes ---**
90.		
…		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]
BarcodePrefixChar=$

If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font "Codedreineun" and prefix "."

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	**Barcode**	**Raw data**
50.	*50.1337*	KNR = 1337

9.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR **BarKenn91=EGR:GUT**	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345“ → SAPCNR=12345

Firmly assigned barcode prefixes:
10:ANR
11:AUNR
12:AFOLG
13:AGNR
14:UAGNR
15:RMNR
16:MNR

17:MST
18:AUSGRD
19:AGGGRD
20:BPOS
21:LPKZ
22:SPLNR
40:ZLO
41:TPE
42:CNR
43:DLL
44:CNR:ALT1
45:CNR:ALT2
46:CNR:ALT3
47:CNR:ALT4
48:CNR:ALT5
49:CNR:ALT6
50:KNR
51:PNR
52:KST
53:FGR

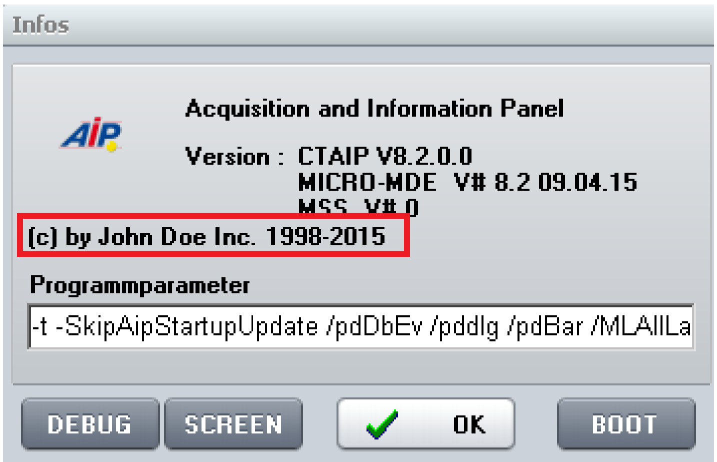
10 AIP2 -Local Configuration

10.1 Local Configuration ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the C:\MPDV\AIP2 directory.

Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

Layout configuration

Entry	Comment
Section [system]	
CompanyInfo=(c) by John Doe Inc. 1998-2015	Overwrites the copyright text in the "About" dialog 
parameters= … -t | +t	The virtual keyboard can be disabled/enabled, irrespective of the terminal type.
DEMO_ALL=ON	The information in the lower bar are normally hidden in demo mode. This option, however, causes all pieces of information to be displayed even in demo mode (version number and date, server, terminal number, online lights).
parameters= … -SkipDynDlgExtraction…	Internal option preventing the extraction of dialog fields/buttons when updating dialog data. (Only applies if WF directory ".\spool\$wf.$$$\" is available)
parameters= … -SkipAipStartupUpdate…	Internal option preventing the configuration files (*.ini,*.cfg,*.cfi) of the module DLLs (packets*.dll) and application DLLs (functions*.dll) from being synchronized when starting the terminal.
VirtScreenSize=640	All windows are started with the indicated resolution

Entry	Comment
VirtScreenRatio=16:9	The display ratio remains if the configuration VirtScreenSize is used to reduce the window. The width-to-height (aspect) ratio can be changed using VirtScreenRatio. Consequently, the width-to-height ratio 16:9 can be tested with a 4:3 monitor and vice versa. The value can be configured as "16:9" or as floating point value (e.g. 1,77777). Note: Information screens are specifically scaled if monitor screens in portrait format (e.g. 9:16) are used. This special scaling can be disabled in hytnrcfg.ini: [Tnr Konfiguration 0]->PortraitAlignment=off
Section [SKIN]	Configurations for the terminal's skin (see ctaiplay.ini) The online configuration of the terminal can be reached by the info dialog (click MPDV icon). ALT+F1 opens the control elements for the skin. Please note: Only in the design/GUI of AIP 8.1 and/or the dynamic dialogs
Saturation=0	Skin configuration / Default = 0
Hue=0	Skin configuration / Default = 0
Name=mpdv	Skin to be used / Default = mpdv
Active=false	Skins can only disabled in ctaip.ini! Default = true

Entry	Comment
Section [system]	
`Usr=21`	Distinct terminal number
`Hostname=192.9.200.24`	Internet address of the server
`Offlinetimeout=600`	The interval after which online access should be attempted the next time. The interval is specified in seconds
`Showcursor=on`	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
`Loadfile= ctnet\win\aip2.txt`	Configuration file to download the application from the server. The paths is relative to the installation directory of the system.
`Watchdog=on`	ON: Watchdog is activated OFF: Watchdog is not activated

Entry	Comment
`Demo=off`	'on': Offline demo mode; always off in the production environment!
`parameters=-t`	The –t parameter switches off the virtual keyboard.
`TMOUT_C=xxx`	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_S=xxx`	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_R=xxx`	Timeout for RECEIVE of the server If not specified, default = 120 seconds
`TMOUT_F=xxx`	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	**Configuration of customized barcode prefixes.**

`BarKenn90=MNR4`
`...`
`BarKenn99=ANR3`
	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.

Section [comports]	

`com1=0`
`com2=MSS`
`com3=BAR`
`Com3=LEGIC`
`Com4=RFLESER`
	Assignment of serial interfaces to connected devices:
	MSS – machine interface
	BAR, LEGIC, RFLESER – various reading devices

Section [MSS-INIT]	
`MSS_DIALOG=10`	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. The dialog closes automatically with "Yes" after an entered time has elapsed; in this case the counting impulses are posted. This value configures the time in seconds the dialog is open. If the terminal is switched off for less than 15 minutes, no dialog is opened; the counting pulses recorded in the switch-off phase are accepted and posted without confirmation. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.

Entry	Comment
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	Additional configuration for **MSS_DIALOG** If the backup file for counting pulses of the terminal is older than 5 minutes, no dialog is opened and the back up file deleted. Quantities recorded at the time when the terminal was closed are not used/posted. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be brought to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: Parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [DLL]	
BusDLL=PCC.EXE	PCC.EXE must be entered here if the terminal is configured with the option "operated as HYDRA-MDE terminal". If necessary, the AIP2 enters this value independently upon starting the program.
Section [PDV]	
PDVProtokoll=ON	Enables PDV logging (prot_pdv.txt) (by default=OFF)
PDVIOPDir=c:\IOPSim\	Path for DNC
Section [CAQ]	
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.
FieldKNRBarcodeOnly=Y	If this entry is set the badge number may only be entered using a scanner.

Entry	Comment
`BarcodeWNR=`	This field specifies which acronym is entered into the tool number field by the scanner.
`BarcodeNest=`	This field specifies which acronym is entered into the cavity number field by the scanner.
`BarcodeNumm=`	This field specifies which acronym is entered into the number field by the scanner.

10.2 PNG – Files / Bitmaps

The use of PNC files is recommended by MPDV. By default PNG files have a size of 24 x 24 px.

10.2.1 File pict.zip

The file "pict.zip“ is updated by the installation tool "inst32.exe“ while downloading and includes all default PNG files.

The default PNG files can be overwritten in the file pict_cust.zip. Several PNG files have the extension ".small.png" (e.g. aip.small.png). These PNG files are used with a screen resolution of 640x480.

10.2.2 File pict_cust.zip

The file "pict_cust.zip“ is loaded from the server directory (e.g. \<serverDir>\1\custom) when starting the program (as is the case for the hycust.mld).

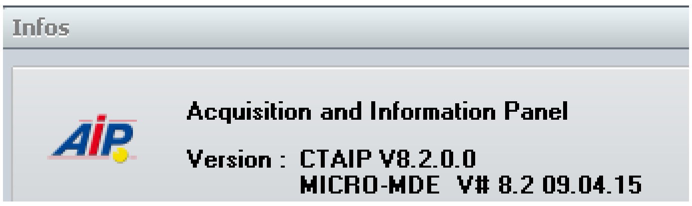
Customized PNG files may be stored in this file and loaded by the AIP2 terminal. Default PNG files may also be "overwritten".

Please note: file sizes are not adjusted.

Customize header

The AIP icon displayed in the header can be replaced by storing a separate AIP.png file in the pict_cust.zip file.

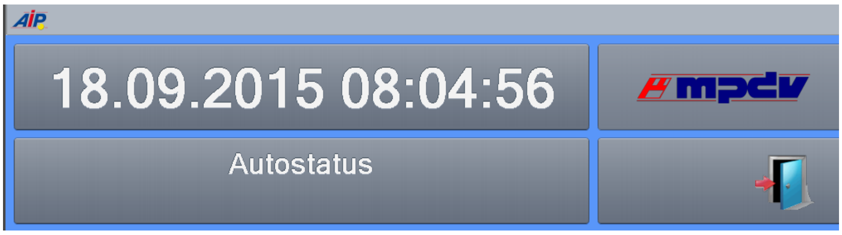
This AIP icon will also be replaced in the "About" dialog.



Customize footer

The MPDV icon displayed in the footer can be replaced by storing a separate company.png file in the pict_cust.zip file.

Customize PZE dialog



The MPDV icon displayed in the PZE dialog can be replaced by storing a separate pze_mpdv.png file in the pict_cust.zip file. In case the PZE terminal is operated with a screen resolution of 640x480, a customized pze_mpdv.small.png file has to be integrated in the pict_cust.zip file.

10.3 Multilingualism (*.mld files)

The below-mentioned files are required for the translation of the application. The table shows the priorities for the translation, ownership and the relevant storage locations.

Priority	File	Server directory	Owner	Description
1	ctaipkd.mld	./ctnet/win/aip2/custom	Customer	Customer-specific translations
2	hycust.mld	./1/custom/	MPDV	Customized translations by MPDV. The file is used by the AIP2, CTAIP and CTWIN terminal.
3	ctaip.mld	./ctnet/win/aip2	MPDV	Standard translation file

11 AIP2 - Central Configuration File hytnrcfg.ini

You can use this file as central place to store different configurations for all or for separate terminals.

For each section, a general version is available
[section 0].

The entries included in this section can be overwritten by entries in a terminal-specific section
[section <TNR-USER>]
<TNR-USER> = UserNo = terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/UserNo

The file hytnrcfg.ini is loaded from the server on every terminal start.

Section / Entry	Comment
	→→→
[Tnr Konfiguration 0]	
`FollowExternStatus=on`	Transfer of machine statuses when reloading machine list. Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
`InstallFonts=on`	If set to "off", fonts are not installed during restart. ON=DEFAULT
`OnlyInstallFontsAfterDownload=false`	If "`InstallFonts=on`": If true, then fonts are only installed directly after a download. If false, then fonts are installed every time the terminal is restarted. (false = DEFAULT)
[Terminal->USR 0]	

Section / Entry	Comment
`AttachedApplication=First`	Displaying documents of OP info: With this configuration, the system first checks whether or not an application is linked in Windows that matches the file extension of the document. This application is then used to display the document. If no link is available, the viewers configured in ctaip.ini (→ [ext. software]) and internal viewers are used. If an extension is completely unknown, the system tries to display the document as text . Different settings are possible: **First** → search for linked application first **AfterUserViewer** → If a UserViewer is configured, this one overrides the linked application (also applies to ExcelViewer, WordViewer and PowerpointViewer) **Last** → Only if no ctaip.ini assignment is found for the file extension, then the system searches for a linked application (**default**). **Off** → The system does not search for a linked application.
`HTTPBrowser=standard`	Display of documents (via OP info): If documents are configured with a path of schema "http", the file is not downloaded to the terminal, but the link is transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows is used.
`SupressErrorMessage=70012`	Suppress message "material is not planned"
`MSS_DIALOG=10`	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. After a configurable period of time, the dialog closes automatically with "Yes" (Yes, posting of pulses). This value configures the time in seconds the dialog is open.
`MSS_FILEAGE_MIN=5` `MSS_FILEAGE_OVERTIME=delete`	If the backup file for counter pulses on the terminal is older than 5 minutes, then no dialog is opened and the backup file deleted. Quantities recorded at the time when the terminal was closed are not used/posted.
[SignatureRecording->User 0]	
`ManualBadgeInput=true`	This configuration specifies whether or not the field *User* can be edited on the terminal (by default: no editing) **true** → activates keyboard input for field *User* on the terminal

Section / Entry	Comment
`Transparency=255`	The display of the signature dialog can also be transparent. **255** → Signature dialog is 0 % transparent (not transparent) **1** → Signature dialog is 99% transparent (maximum transparency) (Default = 155)
`ShowPosition=TR`	You can change the place of the signature dialog: **TL** Top – Left **TM** Top – Middle **TR** Top – Right **ML** Middle – Left **MM** Middle – Middle (Default) **MR** Middle – Right **BL** Bottom – Left **BM** Bottom – Middle **BR** Bottom – Right
`USE_SERVICE_ACCOUNT=1`	**0** (default) SSO: ServiceAccount is not used (requirement: the terminal must be started with the domain "user" (SSO)). **Note:** ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
`SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY`	**REPORTING_USER_READONLY** The user identification using the Windows user is activated. The Windows user is then preassigned in field *User*. The *User* field is read-only. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. **REPORTING_USER_CHANGEABLE** The user identification using the Windows user is activated. The Windows user is then preassigned in field User. The *User* field can be edited. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible.

Section / Entry	Comment
SIGNATURE_1_LOGON_TYPE=HYDRA	**"" / Not set / "EMPTY"** There is also an alternative login procedure. **HYDRA** The user identification using the Windows user is locked. You must enter a user for identification. Requirement: All reporting persons must have been created as users and the option "SSO" must not be enabled for the users. Otherwise, successful authentication is not possible. **ACTIVEDIRECTORY** The TAB for the user identification via *User* is locked. The Windows user must be used for identification purposes. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. **MIXED_BUT_UNIQUE** The setting of option "SSO" specifies whether the user login or the Windows login is used. "SSO" enabled → Windows only "SSO" disabled → user only
SIGNATURE_2_LOGON_TYPE=HYDRA	Identical to SIGNATURE_1_LOGON_TYPE (see above)
ExtendedSignatureRecording=true	Used for signatures on the terminal with quality data collection.
[MDE/Blade Configuration 0]	
CONVERT-TO-ANSI-FILE=<list1|list2>	Configuration of the files that are provided from the AIP to the MDEB2 blade in ANSI format when a combined operation is available. The following lists are transferred by default if the entry is not available. counters.lst|schicht.lst|mnr.lst|mstat.lst|anr.lst|pnr.lst If you want to transfer further lists, you must specify the standard lists and the additional lists.

11.1 Layout configuration

Entry	Comment
Section and/or **[terminal configuration 0]** **[terminal configuration 2XXX];**	**(general configuration)** **(2XXX terminal-specific configuration)**
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	This setting specifies that in case of an error that occurred reading the clock (e.g. when activated after standby

Entry	Comment
	mode), the time is transferred without confirmation dialog and the terminal time is later synchronized with the server time via PDM command.
`SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON`	Suppresses the warning after restart of terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default=OFF)
`CalcTargetYieldSinceLogon=2`	CalcTargetYieldSinceLogon=**1** The duration is calculated from the total runtime since login (all statuses) minus the configured shift breaks. CalcTargetYieldSinceLogon=**2** The duration is calculated from the total runtime since login (all statuses). Defined breaks are not used and are not deducted. (default value)
Section [QRD-PRINTER->TICKET 0] **;(general configuration)** [QRD-PRINTER->TICKET 2xxx] **;(2XXX configuration for a specific terminal)**	
`COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >`	Reloads the machine row for the configured **<Events>**, if it is not available locally ➔ This configuration might be required for a group workplace without machine assignment.
`COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >`	Reloads the order row for the configured **<Events>**, if it is not available locally ➔ This option has been implemented to access order data in the master data, e.g. when logging on orders.

`COMPLETE-..-EVENT=< Events >`
`COMPLETE-..-EVENT=#ALL#`
 	
`COMPLETE-..-EVENT=A_AN|A_P_AN`
	Explanation on the configuration of <Events>
	➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event.
	➔ <A_AN|A_P_AN> restricts reloading of information to specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.

Section **[AIP2 Initialization 0]**	
`XML-GUI=OFF`	Disables the new AIP2 design and uses the AIP 8.1 design.
`CTWIN-STYLE=ON`	Activates the GUI that is similar to CTWIN on the AIP2. The two button bars are shown below the two lists just like on the AIP 8.1.
`CTWIN-BUTTON-LAYOUT=ON`	If the option CTWIN-STYLE=ON is additionally set, the two button bars are displayed at the bottom of the screen.
`AUTOMATIC-CHANGE-TO-START-DISPLAY=30`	As of AIP 8.2.2.28: If this option is set, the display automatically changes to the main view after the configured time if no other interaction was performed in the meantime. The changing display is configured via the option *Show machine/OP* in the *Terminal configuration*, tab *MF functions*. • List: ○ Change from the detail views or function menus (operation, person, resource, etc.)

Entry	Comment
	◦ Change to the main view with "tiles" • Icons: ◦ Change from the detail views or function menus or from the main view with tiles ◦ Change to the icon view of workplaces The configuration is specified in seconds. Do not specify less than 10 seconds. An automatic change to the start screen is not made if an input dialog is open.

12 AIP2 - Local Configuration File keyboard.ini

You configure the virtual keyboard of the AIP2 terminal in the keyboard.ini file in the directory c:\mpdv\aip2 for the specific terminal.

To activate the changes in the configuration file, you must restart the terminal software.

Logic enabling the virtual keyboard:

The AIP2 terminal shows the keyboard if an input field is focused. The keyboard is placed with reference to the field as described below:

Logic for placing the virtual keyboard:

It is tried to place the keyboard directly below the input field. If there is not enough space to the bottom of the screen, it is tried to place the keyboard directly above the input field. If the space above the control element is not sufficient for the keyboard, the keyboard is placed at the bottom of the screen.

These are the priorities for horizontal alignment:

- to the right of the control
- to the left of the control
- to the edge of the screen that is further away from the control

If the “VirtScreenSize“ option is enabled, the virtual keyboard is not aligned on the virtual screen but still on the real screen. Consequently, the keyboard may also reach beyond the terminal program.

The virtual keyboard can be configured in the local keyboard.ini file on the terminal. Example:

```
[Keyboard]
HideTime=10
ScaleMultiplier=0.9
FixNumbers=ON
Configuration=ON
;Logging=ALL
;Processes=ctaip.exe
;ClassesForLetters=TVtEdit
;ClassesForNumbers=TMPDVSimpleNumericField
```

HideTime

The set value specifies for how many seconds the keyboard is invisible if you click on the key showing the icon  on the left hand side. This key is not visible if the value "0" is entered.

ScaleMultiplier

The keyboard size can be reduced and increased. The value range is between 0.9 and 4.0. A dot is used as decimal separator.

The default value is 1.0.

FixNumbers

Allowed values: ON|OFF

If FixNumbers=On is set, the number keys located in the top row of the virtual keyboard remain visible even if the Shift key or CapsLock key is pressed. ON is set by default.

Configuration

Allowed values: ON|OFF

The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard. You can activate different keyboards in the operating system. For the virtual keyboard, you can then switch between the different activated keyboards.

The entry *Configuration=ON* activates the button . Use this button to open the dialog to select one of the keyboards activated in the operating system.

Default is OFF.

Logging

Allowed values: OFF|ON|ALL

Logging can be enabled using this entry. The advanced logging is configured by setting ALL.

OFF is set by default.

Processes

The entry "Processes" specifies for which additional processes the virtual keyboard will be used. The separate entries are separated by comma (e.g. processes=notepad.exe.explorer.exe). If this entry is not included, the keyboard for these processes is available in *ctaip.exe* und *iniedit.exe.*

ClassesForLetters

This entry defines for which additional classes the alpha-numeric keyboard should be displayed. The current classes for AIP2 (TMPDVSimpleField, TsEdit, TsMemo, TMPDVTypEdit, TMPDVSimpleEditField, TMPDVPictureField,TEdit, TButtonedEdit, TEditControl) are fixed in the source code. This entry can be used to extend the list.

ClassesForNumbers

This entry defines for which additional classes the numeric keyboard should be displayed. The current classes applicable for the AIP2 (TMPDVNumericField, TPagerNumField, TMPDVSimpleNumericField, TVTEdit) are fixed in the source code. This entry can be used to extend the list.

The classes that are fixed in the source code cannot be overridden using a different entry in the configuration file. If you want to display the other keyboard for a field, you can change the input type of the field in Dialog Configuration.

Dialog-specific configuration

There is the option from version 1.6.0.0 of the keyboard.exe to configure the location of the virtual keyboard per dynamic dialog. The user has to extend the configuration file keyboard.ini accordingly.

Sample configuration:

[WF_AA_QUA]	=> Name of the dynamic dialogs
X-Position=50	=> Distance in pixels from the left edge of the screen
Y-Position=50	=> Distance in pixels from the top edge of the screen

- Specifying the X- and Y-position is mandatory.
- The configuration is only available for dynamic dialogs that are configured on the MOC.

The virtual keyboard can also be switched off if the terminal is connected to a real keyboard. This can be configured in section [SYSTEM] of the local ctaip.ini file.
Example:

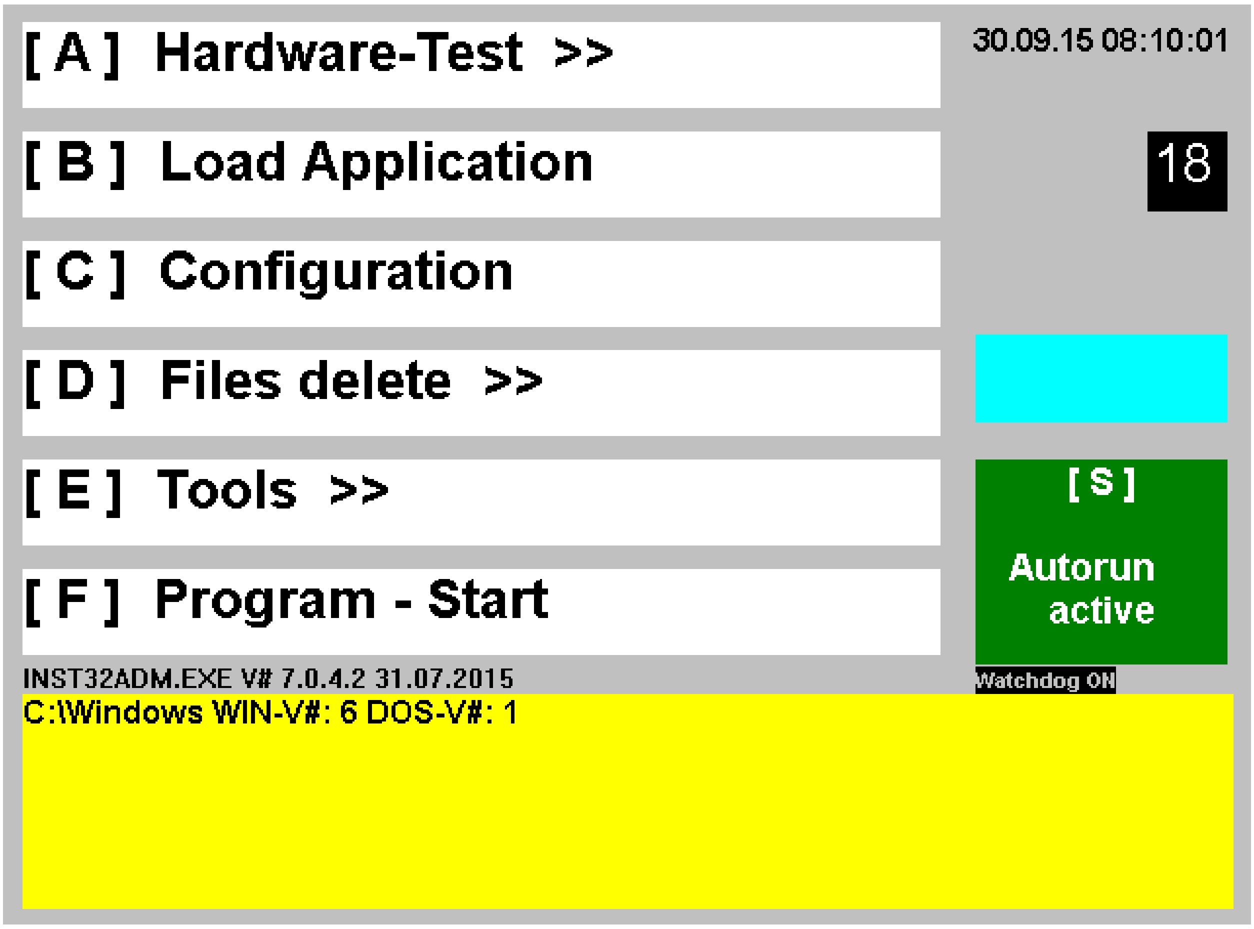
```
[SYSTEM]
Parameters=-t
```

Syntax:

+t/-t --> enables/disables the virtual keyboard; irrespective of the terminal type

13 Start Menu Inst32

The terminal software is installed on a local hard drive of the terminal or a shop floor PC. After booting the terminal, the following selection menu appears:



The following selection options are available in the selection menu (start by key or clicking)

[A] Hardware Test >>

Submenu for different hardware tests

[B] Load Application

After an update from the HYDRA server, the application programs can be easily updated.

[C] Configuration

[D] Delete Files >>

Submenu for deleting different files (data directories, queues, logs, application)

[E] Tools >>

Submenu for opening or installing different tools

[F]   Program – Start

Opening the application program. If nothing is entered, it will be started automatically after 20 seconds.

The **<D>** and **<E>** buttons are only used for service purposes.

If no button is pressed, the AIP2 application starts automatically after 20 seconds. Once the application has started, a few notes are displayed indicating the version number, hardware configuration and other service information. If everything (configuration and terminal) is in order, the display switches to the basic dialogs described in more detail in the sections that follow. If the terminal detects hardware or configuration errors, it alerts the user by sending messages in plain text.

Submenu for opening or installing further [A] Hardware Tests   >>

[1]   MSS Test

(Hardware test for the machine interface / MSS)

[2]   Reader Test

(Hardware test for different reader connections / BAR, LEGIC, PLG, ...)

[3]   LAN Test

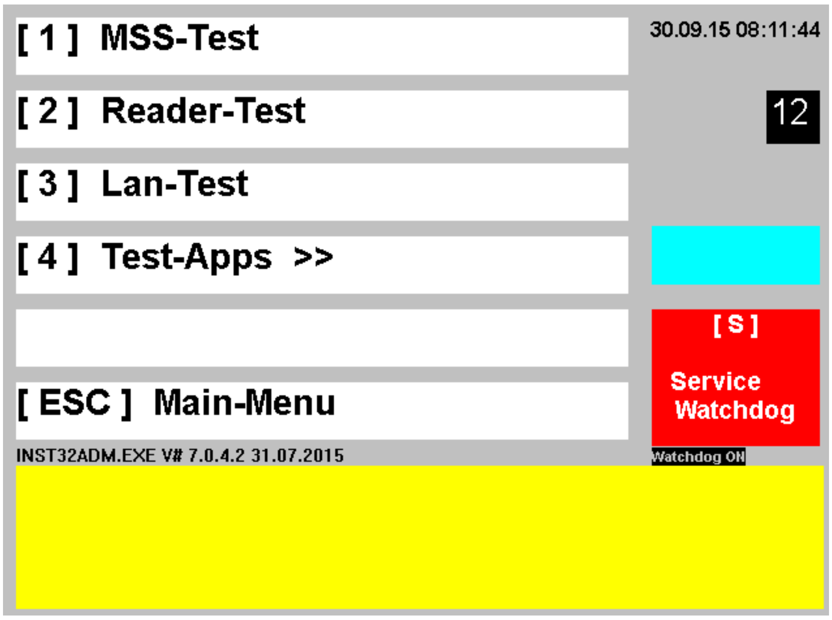
(Hardware test of the network connection)

[4]   Test Apps >>

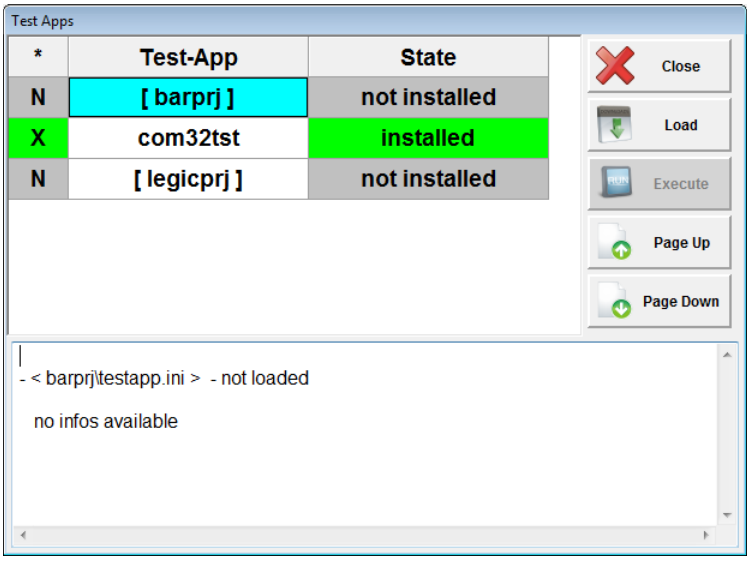
(Test applications of any kind)

[ESC] Main Menu

Back to the main menu



Starting the function “Test Apps“ by the menu item ”[4] Test Apps >>"



When starting the function “test apps“, a directory list of the HYDRA server directory ”./ctnet/win/testapps/“ is loaded.

The function creates an entry for each directory in the list that includes a file “testapp.ini”. If this file already exists locally, i.e. it has already been loaded earlier, the entry will be highlighted in green (e.g. 

).

The selected *“Testapp“* can be reloaded or updated by the function key **”Load“.**

The selected *“Testapp“* is started by the function key **“Execute".**

The list can be browsed by using the function keys **“Page Up”** and **“Page Down”.** The function is exited by using the key **“Close**”.

The configuration file “testapp.ini“ is structured a follows.

```
[APP]
name=com32tst
exe=com32tst.exe
param=...          ; optional parameter to transfer call data
[COMMENT]

      MSS Test Program

[/COMMENT] [APP]
```

Submenu deleting or restarting/closing files using [D] Delete files

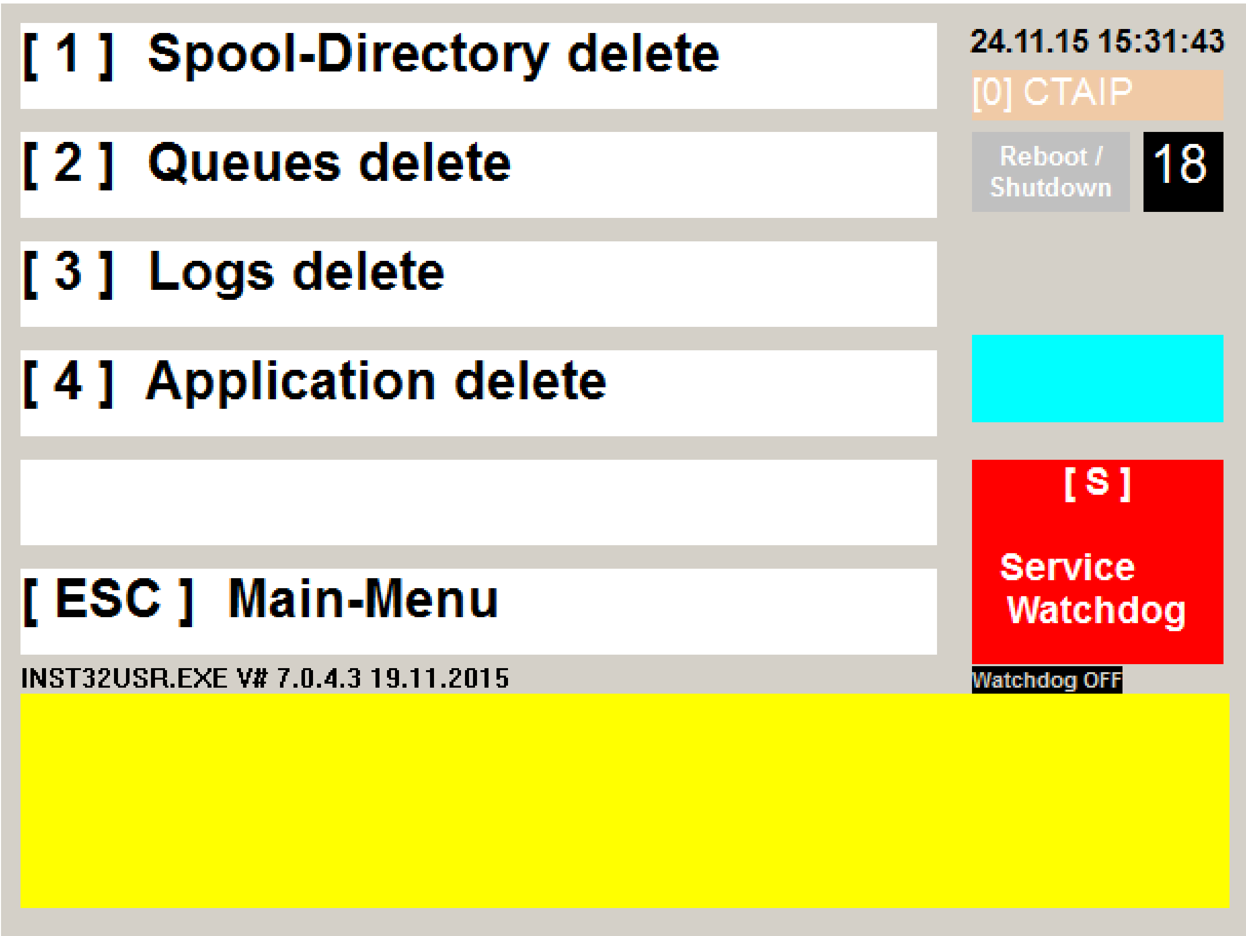
[1] **Delete spool directory**
(deleting the ./spool directory of the terminal)

[2] **Delete queues**
(deleting the queue in the ./spool directory of the terminal)

[3] **Delete logs**
(deleting the log files in the ./spool directory of the terminal)

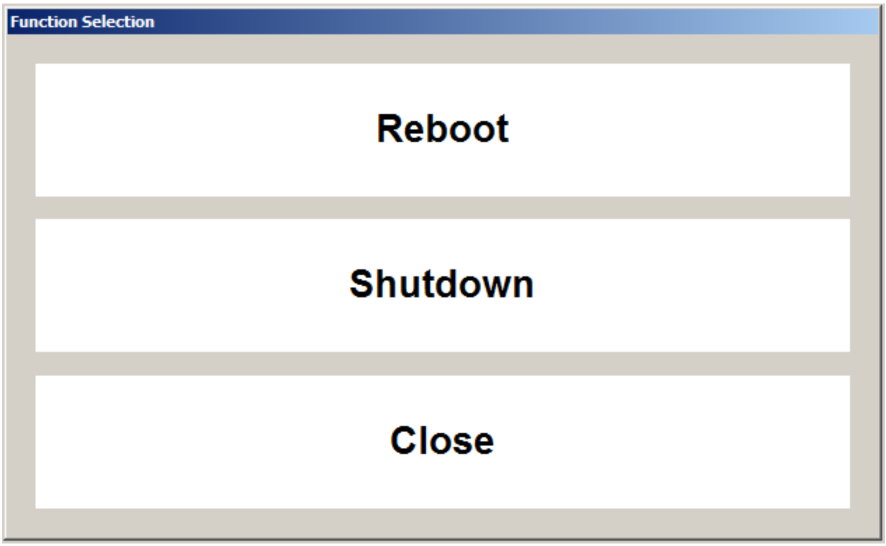
[4] **Delete application**
(deleting the configured application)

[ESC] **Main menu**
back to the main menu



The application / Windows can be exited or restarted in the following selection dialog after entering the password "mos6950" and clicking the button 

.



Submenu for opening or installing further “[E] tools >>“

[1] **MSS-Loader install**

[2] **MSS-Loader start**

[3] **VNC install**

[4] **HYDRA Fonts install**

(loading and starting the installation package for Font Installation)

[5] **more Tools >>**

(Backup, Restore, Terminal Upload, Install 3rdParty)

[ESC] **Main Menu**

Back to main menu

Submenu by starting "[5] more tools”

[1] **Backup**

[2] **Restore**

[3] **Installation 3rd Party**

[ESC] **Main Menu**

Back to the main menu

Menu item "[1] Backup“

The server uses the file **<hydradir>**\ctnet\win\aip2backup.txt
or a terminal-specific file **<hydradir>**\ctnet\win\aip2backup2xxx.txt
(xxx is terminal number) for the backup. At first the system attempts to load a terminal-specific file.

If no terminal-specific file exists, the system will then attempt to load the file aip2backup.txt. This file contains all of the files or registry entries that need to be backed up.

```
\aip2\*.ini
\aip2\*.cfi
\aip2\*.cfg
HKEY_LOCAL_MACHINE\SOFTWARE\MPDV\
```

A Zip file is created in the terminal and it is stored in the server.
The file is located in the server at:
The backup Zip file is given the name: aip2backup2xxx.zip   ->xxx = terminal number   (terminal-specific for Hydra user 2xxx)
This backup file is then stored in the server under

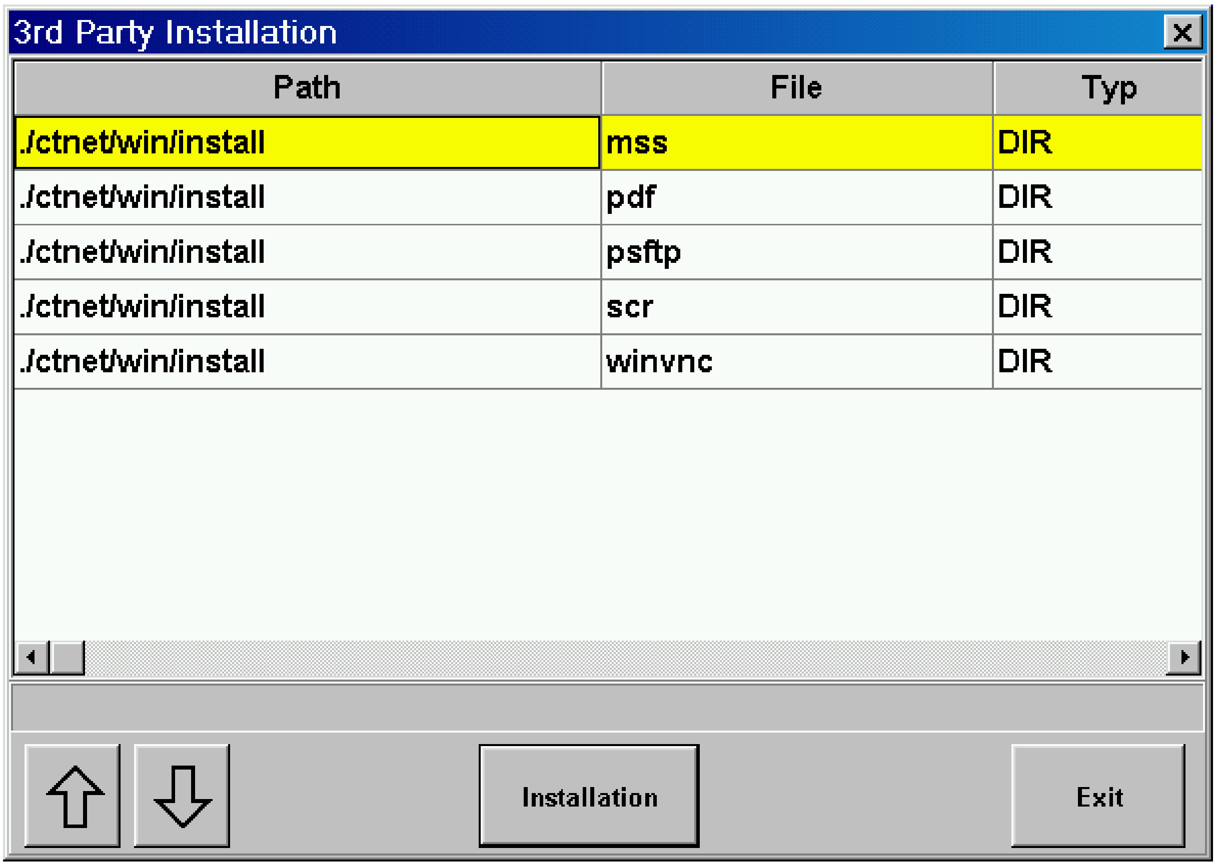
<hydradir>\custom\backup\aip2\aip2backup2xxx.zip .

Menu item "[2] Restore“

There will first be a query asking whether you would like to run a restore. "Restore" attempts to load a backup file located in the server and then automatically restores all of the backed up files and any backed up registry entries.
As already described in the backup section, a backup file is filed in the server directory:

<hydradir>\custom\backup\aip2\aip2backup2xxx.zip

Menu item “[3] Installation 3rd Party“

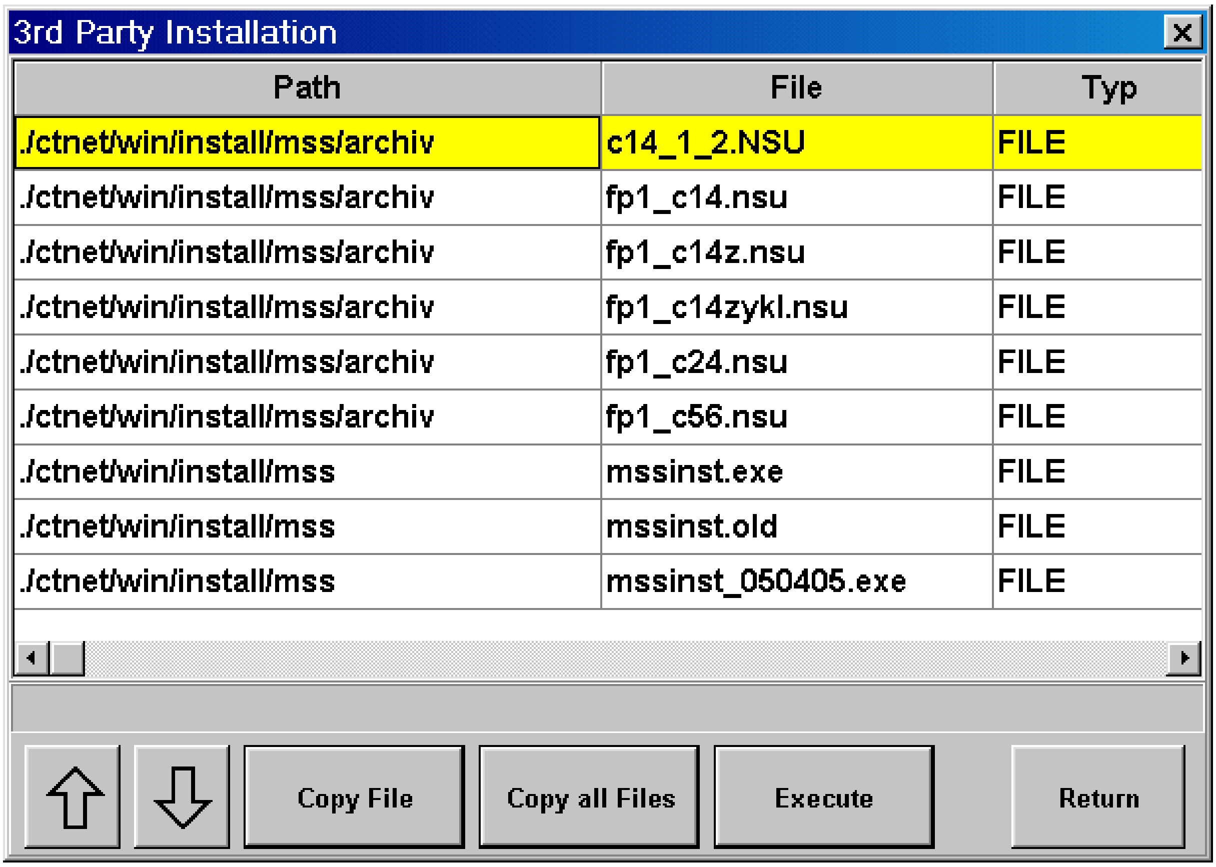


"Installation" button:

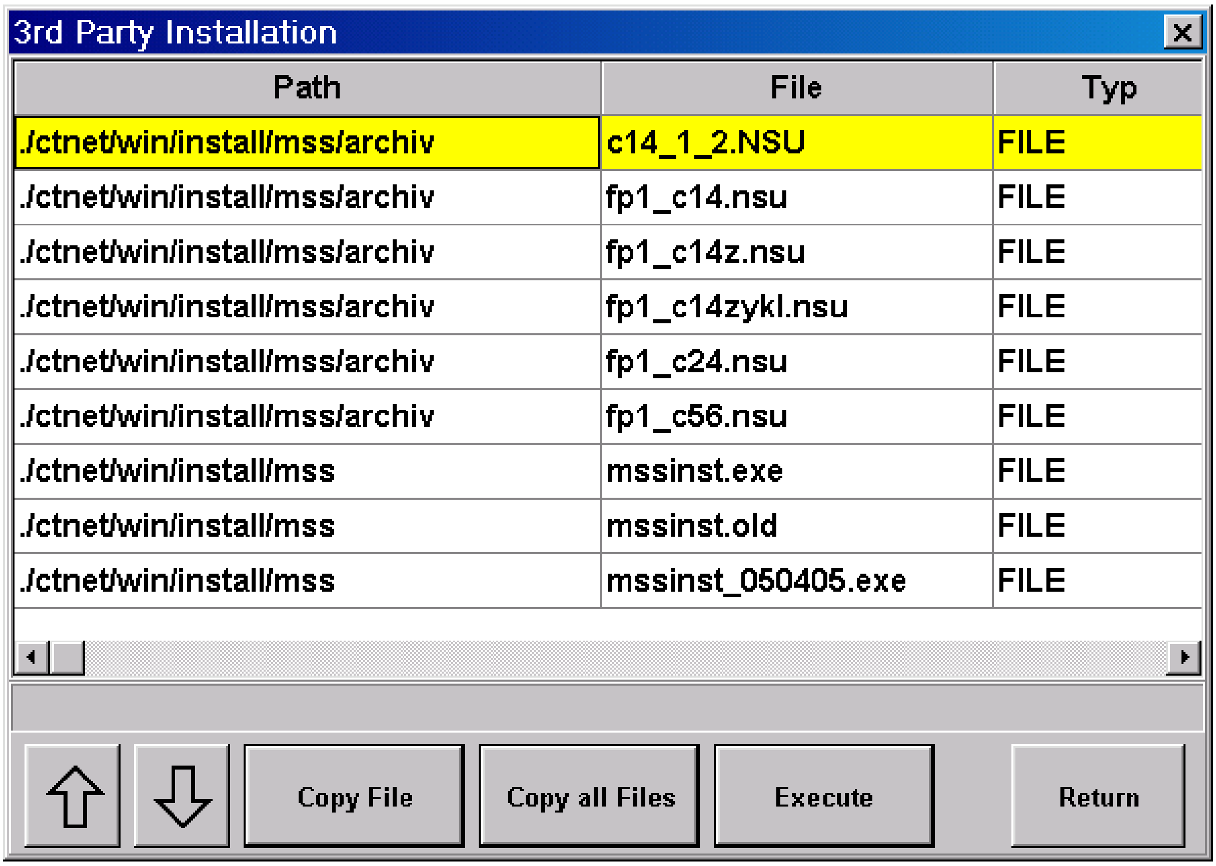
A list with all directories starting with the **<hydradir>\ctnet\win\install** directory is displayed. The directories found will be offered for selection in a dialog. By confirming a directory, this directory is downloaded and its contents are shown.

The **<hydradir>\ctnet\win\install** directory can therefore be expanded to include additional directories.

Content of a directory selected beforehand



Having clicked one of the buttons **"copy file"** or **"copy all files",** a selection screen opens where a directory may be chosen. The selected file or all files displayed are copied into this directory. In order to copy a single file, first it needs to be selected in the list.



***Execute* button:** A file from the list may be executed by clicking this button. The execution program defined in Windows is used to display or execute the selected file.

13.1 Using the functions in Windows 7

Availability or feasibility of individual functions of the “INST32” installation function using Windows 7 depends on the user type:

User Type **INST32 Function**	Local User Member of the “administrators“ group	Local administrator	Local User, Member of the “users“ group
Updating INST32 from the HYDRA server	Yes	Yes	Yes
Installation of fonts	Yes	Yes	No
Loading the application (download)	Yes	Yes	Yes
Changing configuration settings	Yes	Yes	Yes
Installation MSS Loader	Yes	Yes	No
Installation VNC	Yes	Yes	No
Backup	Yes	Yes	Yes
Restore	Yes	Yes	Yes
Installation 3rd party components	Yes	Yes	No

User Type / INST32 Function	Local User Member of the “administrators“ group	Local administrator	Local User, Member of the “users“ group
Deleting files	Yes	Yes	Yes
Starting MSS Loader	Yes	Yes	No
Starting hardware test	Yes	Yes	No
Starting reader test	Yes	Yes	No
Starting LAN test	Yes	Yes	Yes

13.2 Installation of font types in Windows 7

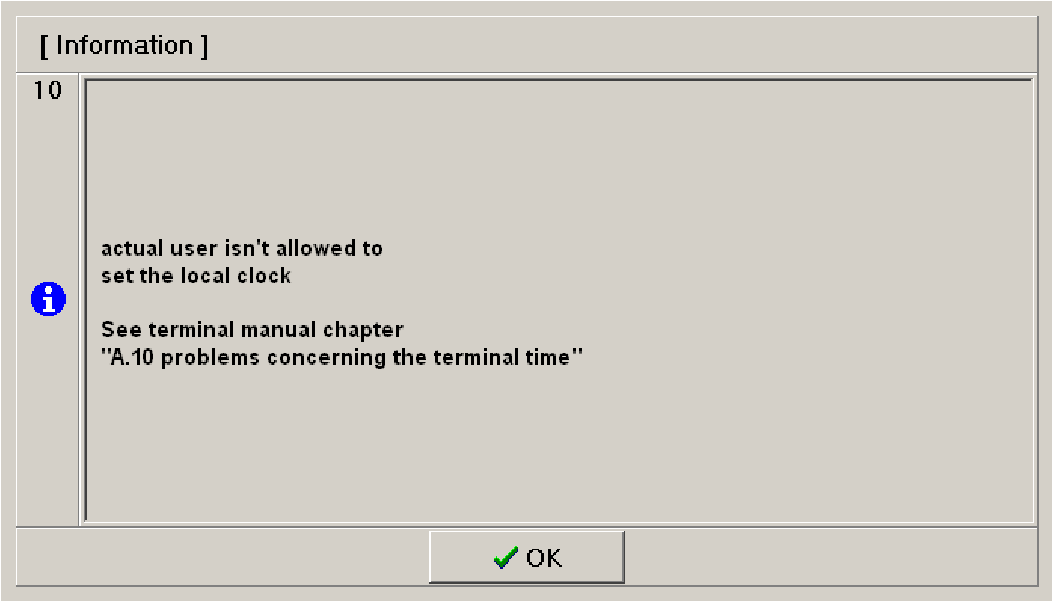
If the "Windows 7" operating system is used the required font types can no longer be installed by the application itself.

The required fonts have to be installed once using the above-mentioned installation program "INST32", menu item "HYDRA Fonts install". This is required to make sure the information is properly displayed on the terminal.

13.3 Date/time synchronization at the terminal

The HYDRA terminal software automatically synchronizes the time of the local terminal PC with the time of the HYDRA server. Usually, different Windows versions require administrative rights to be able to set the time locally.

The message below is displayed once, every time the program is started if the respective Windows user does not have the required rights.



The following entry in the [system] section of the local configuration file CTAIP.INI prevents the time from being synchronized with the HYDRA server and, as a result, the above-mentioned error message is also suppressed.

```
[system]
NoLocalWatchUpdate=on
NoLocalWatchWarning=on
```

But if this configuration is enabled, the user has to make sure that the time of the HYDRA terminal PC is synchronized with another reliable source and that the time of the HYDRA terminal PC and the time of the HYDRA server are synchronous.

The user is responsible for disabling the time synchronization function. Wrong time stamps at the terminal may lead to unintentional problems when processing data, e.g. incomplete or wrong reports and evaluations or errors in posting times and durations.

13.4 Control of special watchdog hardware

New watchdog types are connected by a driver DLL, which is configured in ctaip.ini.

[DLL]
WdDLL=<Driver DLL>

Watchdog	Configuration	Required files
AEC6810	WdDLL= aaeondrv.dll	aaeondrv.dll aaeonWdt.dll

Watchdog	Configuration	Required files
		aaeonwrapper.dll

A watchdog may also be activated within the registry. The following entry has to be set:

HKEY_LOCAL_MACHINE\SOFTWARE\MPDV\CT\WdDLL=<Driver DLL>

In case both entries are set, the entry in the ctaip.ini file takes priority.

13.5 inst32.ini - Standard

Default settings for installing AIP2.

Entry	Comment
Section [install]	
`PrgIniFile=C:\MPDV\AIP2\ctaip.ini`	Path of the ctaip.ini file
`PrgExeFile= C:\MPDV\AIP2\ctaip`	AIP2 program to be started.
`DisplayName=AIP2`	The option DisplayName may only be used with an AIP2 terminal. The value AIP2 must not be changed.
`ConfigEditorFile=aip2.mkf`	Configuration file for the configuration editor. This file controls the GUI of the configuration editor.
`ConfigHelpFile=iniedit.ini`	Help file for the configuration editor.

The default settings should be sufficient in the inst32.ini file.

13.6 inst32.ini - application selection

The following optional configuration can be used if a terminal PC must support different installations during a transitional period (e.g. version upgrade from CTAIP to AIP2).

This configuration should only be used in exceptional cases.

Entry	Comment
Section [install]	
`ApplicationChoiceAvailable=on`	The application to be supported can be selected using this optional configuration.

Single "install*" sections for different applications must be configured with the above-mentioned configuration option.

Section for AIP2

Entry	Comment
Section [installaip2]	
`PrgIniFile=C:\MPDV\AIP2\ctaip.ini`	Path of the ctaip.ini file
`PrgExeFile= C:\MPDV\AIP2\ctaip`	AIP2 program to be started.
`DisplayName=AIP2`	The option DisplayName may only be used with an AIP2 terminal. The value AIP2 must not be changed.
`ConfigEditorFile=aip2.mkf`	Configuration file for the configuration editor. This file controls the GUI of the configuration editor.
`ConfigHelpFile=iniedit.ini`	Help file for the configuration editor.

Section for CTAIP

Entry	Comment
Section [installaip]	
`PrgIniFile=C:ctaip\ctaip.ini`	Path of the ctaip.ini file
`PrgExeFile=C:\ctaip\ctaip`	CTAIP program to be started.

Section for CTWIN

Entry	Comment
Section [installctwin]	
`PrgIniFile=C:\ctwin\ctaip.ini`	Path of the ctwin.ini file
`PrgExeFile= C:\ctwin\ctwin`	CTWIN program to be started.

Default application

Entry	Comment
Section [system]	
`Default=installaip2`	When starting inst32, AIP2 is supported by default.

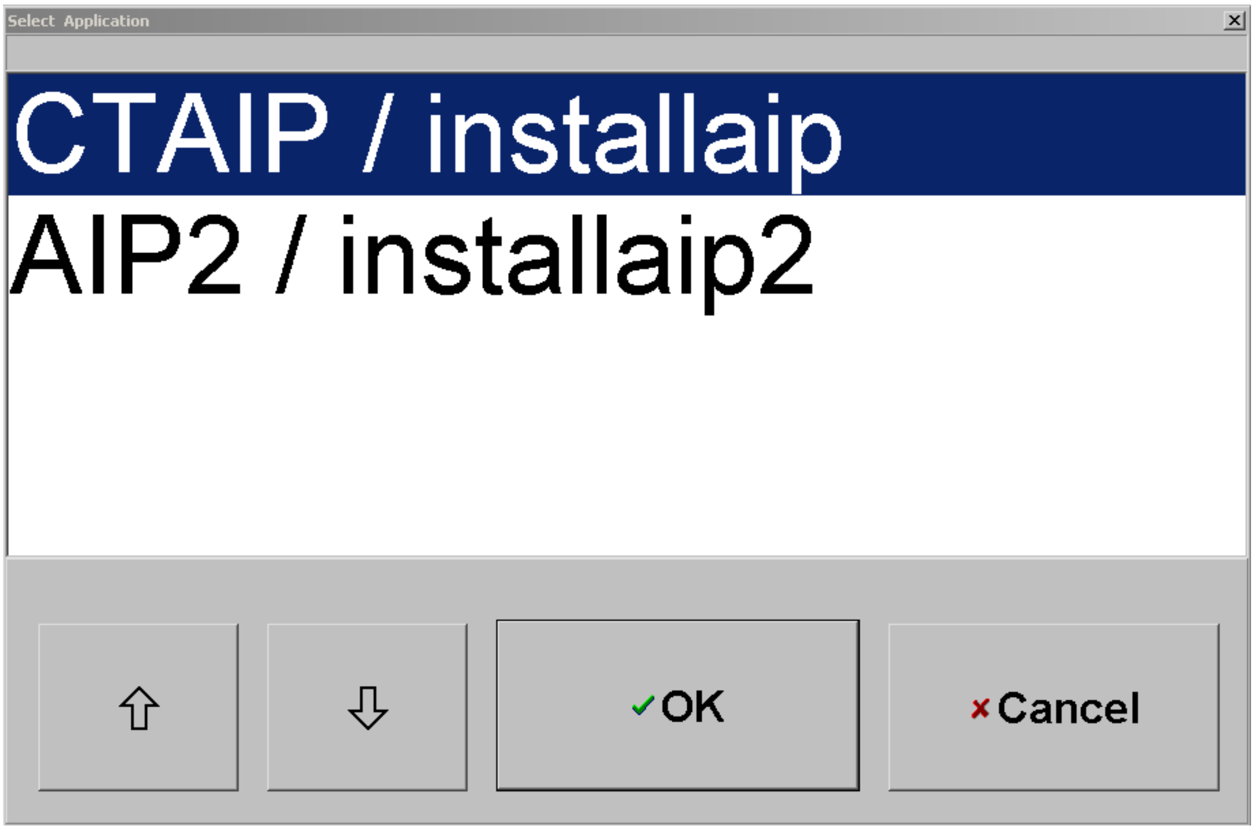
Sample configuration inst32.ini for AIP2/AIP:

The AIP2 has been designed to be configured by default. The AIP2 is selected by default when starting inst32.exe. The application can be changed in inst32adm.

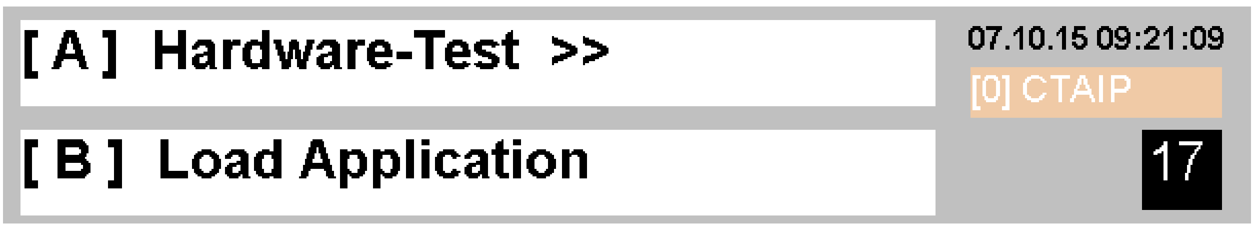
```
[system]
Default=installaip2

[installaip2]
PrgIniFile=C:\MPDV\AIP2\ctaip.ini
PrgExeFile=C:\MPDV\AIP2\ctaip
DisplayName=AIP2
ConfigEditorFile=aip2.mkf
ConfigHelpFile=iniedit.ini

[installaip]
PrgIniFile=\ctaip\ctaip.ini
PrgExeFile=\ctaip\ctaip


[install]
ApplicationChoiceAvailable=on
```

Using this configuration the GUI shows the supported application. By double clicking, the selection dialog can be opened in inst32adm.

The application can be changed in the selection dialog.



Inst32 now supports CTAIP.



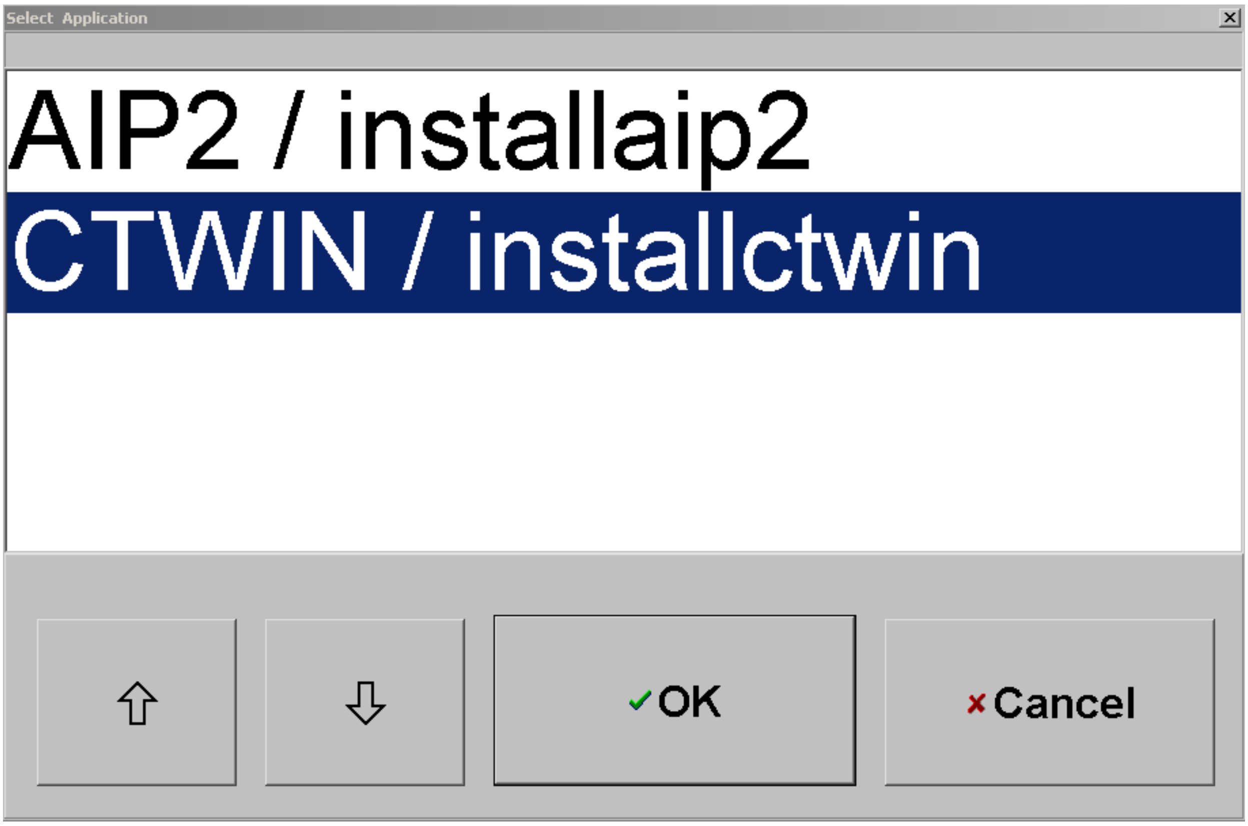
Sample configuration inst32.ini for AIP2/CTWIN:

The AIP2 has been designed to be configured by default. The AIP2 is selected by default when starting inst32.exe. The application can be changed in inst32adm.

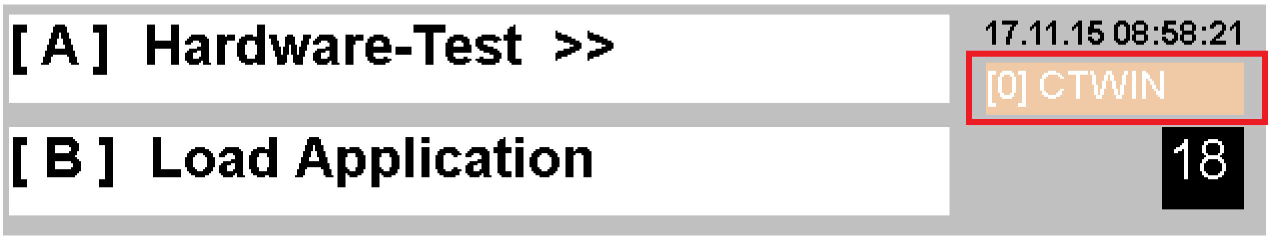
```
[system]
Default=installaip2

[installaip2]
PrgIniFile=C:\MPDV\AIP2\ctaip.ini
PrgExeFile=C:\MPDV\AIP2\ctaip
DisplayName=AIP2
ConfigEditorFile=aip2.mkf
ConfigHelpFile=iniedit.ini

[installctwin]
PrgIniFile=\ctwin\ctwin.ini
PrgExeFile=\ctwin\ctwin


[install]
ApplicationChoiceAvailable=on
```

Using this configuration the GUI shows the supported application. By double clicking, the selection dialog can be opened in inst32adm.

The application can be changed in the selection dialog.



Inst32 now supports CTWIN.

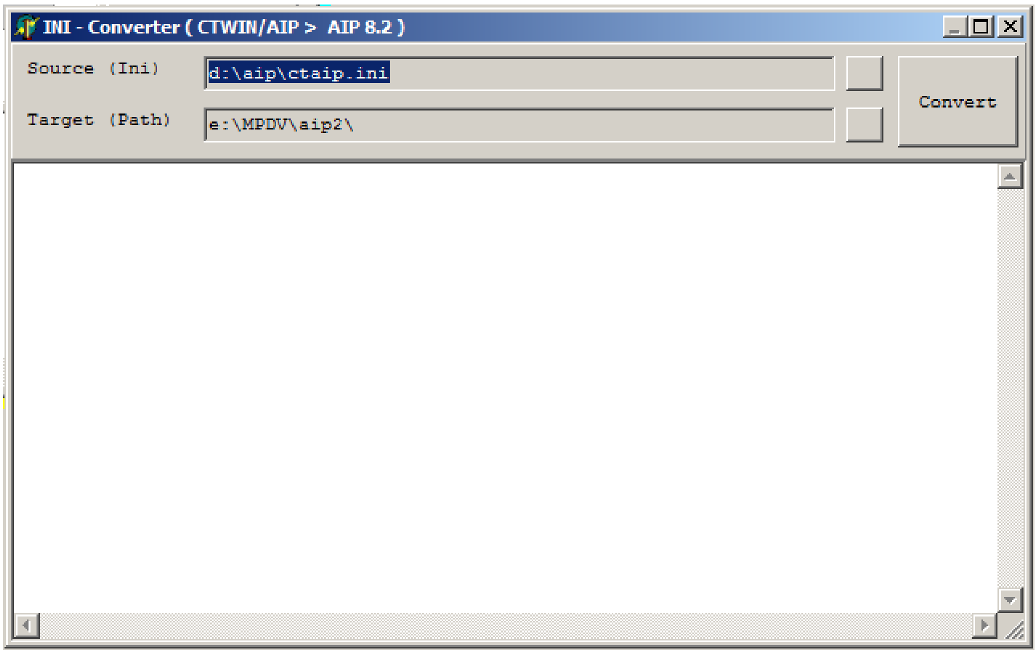


13.7 Migration: CTWIN/AIP --> AIP 8.2

The program *"iniconverter.exe"* is available in order to migrate a CTWIN/AIP installation to an AIP 8.2 installation. This program transfers relevant INI files or INI entries. This program can be loaded and started via the menu item "**`[A] Hardware test >>`**" of the sub-menu "**`[ 4 ] Test Apps >>`**" and by selecting "**`[ iniconverter ]`**".

A detailed description on how to operate "test apps" can be found in the chapter *"Start menu Inst32"* of section *"Starting the function "test apps" using menu item "[4] Test apps >>".*

The following application dialog is shown after loading and starting the test app **`"[ iniconverter ]".`**



While starting the application, the required entries in *"Source (INI)"* and *"Target (Path)"* are assigned by default.

The following order applies for *"Source (INI)":*

- c:\aip\ctaip.ini
- d:\aip\ctaip.ini
- c:\ctwin\ctwin.ini
- d:\ctwin\ctwin.ini
- Registry "HKEY_LOCAL_MACHINE\SOFTWARE\[*Wow6432Node*\]Mpdv\CT\PATH\AIP"
- Registry „HKEY_LOCAL_MACHINE\SOFTWARE\[*Wow6432Node*\]Mpdv\CT\PATH\CTWIN"

The *"Target (Path)"* is pre-assigned via the following registry entry:

- Registry "HKEY_LOCAL_MACHINE\SOFTWARE\[*Wow6432Node*\]Mpdv\CT\PATH\AIP2"

If the values assigned by default are incorrect, they can be revised using the buttons behind the input field.

The *"Source (INI)"* field can be changed using the standard dialog "open file".

The *"Target (Path)"* field can be changed using the standard dialog "search folder".

The button Convert converts/transfers INI files from the *"Source"* directory to the *"Target"* directory.

This command can be executed several times. In case the *"Target"* directory already includes several INI files, a dialog opens where updating/overwriting of these files must be confirmed. *(Please note: all dialogs can be confirmed provided that the AIP 8.2 terminal has not been implemented manually beforehand)*

The transfer result is documented in a memo field and stored in the file "iniconverter.txt" of the *"Target"* directory.

```
Started: 2015-12-01 12:02:09

 Path : d:\aip\ >>> e:\MPDV\aip2\
 +++ update < ctaip.ini > from < ctaip.ini >
 >>> ctaip.ini <0> ==================================================================
 >>> transfer section < System >  to e:\MPDV\aip2\ctaip.ini =========================
 --- < hostname=win2008-7 >
 --- < usr=241 >
 --- < UpdateTime=1 >
 --- < MaxProtSize=4194304 >
 +++ set < hypath=./ > (CONST)
 +++ set < loadfile=ctnet\win\aip2.txt > (AIP 8.2)
 +++ set < backupfile=ctnet\win\aip2backup.txt > (AIP 8.2)
 >>> transfer section < Comports >  to e:\MPDV\aip2\ctaip.ini =======================
 --- < com15=LEGIC >
 --- < com16=BAR >
 --- < com1=MASTER >
 --- < com2=MSS >
 +++ set < com1=0 > (remove MASTER)
```

<u>*Please note:*</u>

- The configurations of PCC drivers, such as those included in the files mssmpdv.ini, opcmdv.ini, etc.
  must be checked and adjusted to the new installation, if necessary.
- The INI files used for the automatic transfer are filed together with the file "iniconverter.txt" in the sub-directory ".\ini-srce\<yyyymmdd-hhmmss>\." of the *"Target"* directory.

14 Configuring the machine image for the AIP

Purpose

You can display the image of the workplace in the AIP shop floor software (in the following "machine image"):

- In the machine info (AIP 8.1 and AIP 8.2)
- In the main view (only AIP 8.2).

Requirements

The following image formats are supported: jpg, gif, png, tif, bmp, ico, emf, and wmf. File the images (files) in a directory that may be accessed from the AIP terminal via the path ID "HYDRA" within the path configuration.

Log your Windows user on to the AIP and access the directory. Your user must have the respective authorization to access the directory.

Procedure

1. Store the file to be displayed in a central directory that you can access from the AIP. Make sure that the Windows user in the AIP has the respective authorization to access the directory.
2. Configure the logical path "HYDRA" in the Path configuration.
   - Path: "HYDRA" (cannot be changed)
   - Protocol: "file" (cannot be changed)
   - Host: IP address or host name of the server where the graphic files are stored.
   - Port: usually 0
   - URL path: Enter the URL path as absolute path. Refer to the network directory where the graphic files are stored.
     In case of a HYDRA server, store the files in the directory <HYDRADIR>/<system>/grafik/bde (<HYDRADIR> is the directory where HYDRA is installed, <system> is the HYDRA system number).

     Precede the URL path by a double backslash.

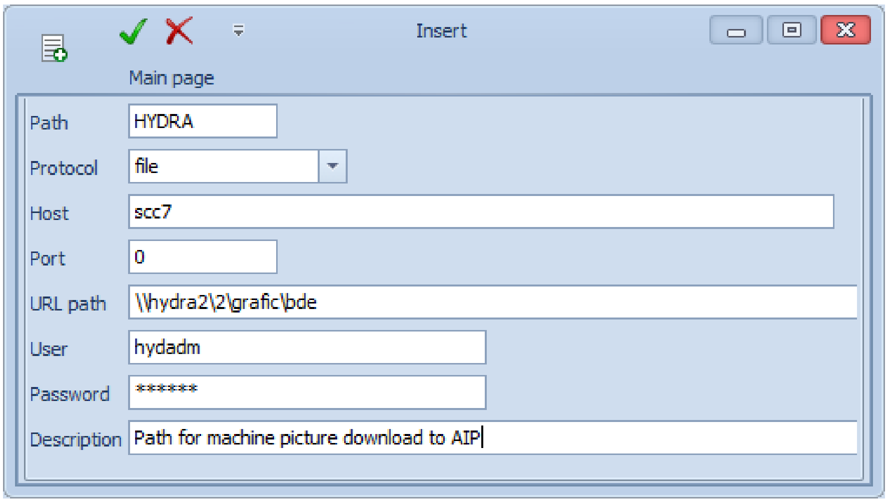
   - User/password: Enter the user and the password of the Windows user to access the server. In case of a HYDRA server, it is usually the user *hydadm.*

   Sample path configuration

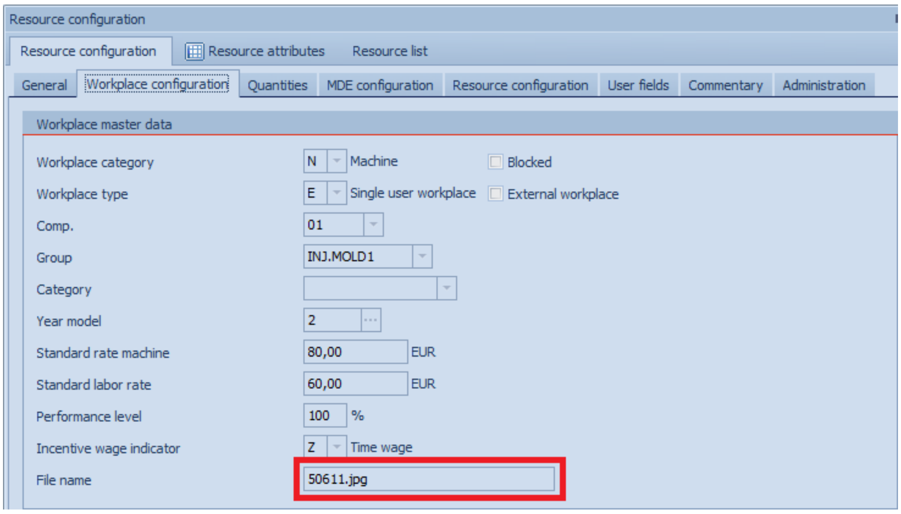
   The graphic files are stored in the directory of the HYDRA server: d:\hydra2\2\grafik\bde

Network share for the directory d:\hydra2 in the HYDRA server: hydra2

Configuration:



3. Enter a valid file name in the field "File name" of the application [Workplace and resource configuration]() (tab Workplace configuration).

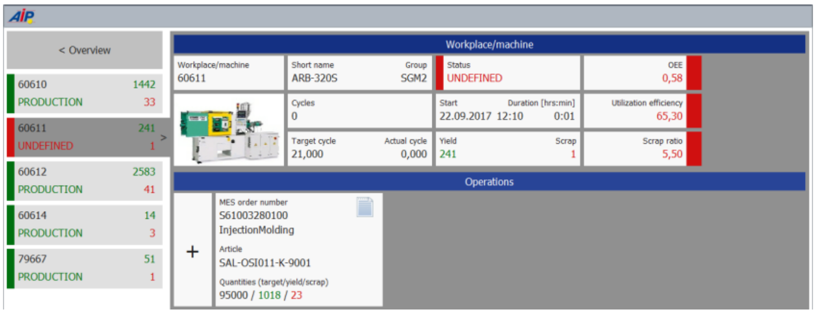


4. Restart the terminal software.

Result

The machine info in the AIP displays the machine image.

With AIP 8.2, the main view displays the machine image:



Troubleshooting

If the main view of the AIP 8.2 does not display the machine image after restart, restart the AIP 8.2 a second time.

If the image is still not displayed, try the following:

- The following entry must be available in the log file prot_ev.txt, e.g.
  `16-10-16 13:32:57.763[+Loc]12260:URLDownload:`
  `file,hydadm,hydadm,SCC7,0,\\hydra2\2\grafik\bde\60610.jpg``,c:\ctaip\spool\60610.jpg,`
  `16-10-16 13:32:59.320[+Loc]12260:URLDownload: =>` **`Res=0`**
  In the row "URLDownload", the return code is displayed that the communication software has returned during download. The return code must be 0 (**`Res=0`**). The log file is written in the subdirectory c:\ctaip\spool (if the terminal software is installed in c:\ctaip).

  The logging of the log file must be *explicitly* enabled in the terminal software. Only then, the entry is displayed in the log file prot_ev.txt: ☑ **DB-Messages (prot_ev.txt)**

- You must have entered an absolute URL path in the MOC path configuration.
- The user that accesses the directory of the HYDRA server from the terminal software must have sufficient (read) access to this directory.

Only one image is downloaded from the HYDRA server. If you want to change the image, you must first delete the file in the spool directory of the terminal. Or you can enter the image using a different name in the Workplace/resource configuration.